



Stochastic dynamics of the resistively shunted superconducting tunnel junction system under the impact of thermal fluctuations

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ABSTRACT

In this work, a Josephson junction consisting of two superconducting layers sandwiching an insulating layer is explored, which is subject to the effects of thermal fluctuations. A clockwise hysteresis cycle in the current–voltage characteristic curve is demonstrated. Additionally, the bifurcation of a planar limit cycle is established. The numerous stochastic thermodynamic properties of the resistively shunted superconducting tunnel junction system are described, considering the influence of three specific parameters: conductance, current bias and noise intensity. Moreover, the probability density is characterized using the Fokker–Planck equation. Crucially, simulations reveal a regime of intermediate thermal noise intensities that maximizes metastable state lifetimes. This noise-enhanced stability effect, characterized by non-monotonic mean first passage time profiles, demonstrates how noise can stabilize or accelerate barrier escape.

1. Introduction

The Josephson effect [1] is a phenomenon that occurs when two superconductors are placed in proximity, with some barrier or restriction between them. This effect produces a current known as a supercurrent, which flows continuously without any applied voltage across a device called a Josephson junction. These junctions consist of two or more superconductors coupled by a weak link that can be a thin insulating barrier or another type of restriction. Rey et al. [2] explored the physical phenomena occurring in such junctions, particularly the interplay between Andreev reflection and Coulomb blockade. They highlighted the importance of understanding these quantum mechanical effects in the context of superconducting junctions, which are crucial for the design and operation of devices like superconducting single-electron transistors.

Barone and Paternò [3] presented a comprehensive exploration of the Josephson effect, encompassing both the fundamental physical principles and the current and prospective engineering applications at that time. In this analysis, they delved into various macroscopic quantum phenomena that held significant potential for technological advancements. Following the theoretical foundation, relevant discussions on the utilization of these effects in devices and their practical implications were also included. Likharev [4] provided an in-depth examination of the statistical, dynamic, and static characteristics of Josephson junctions. Particular emphasis was placed on the dynamics of new circuits and analog and digital devices utilizing single quanta of magnetic flux. Tinkham [5] examined various aspects of

the Josephson effect, in terms of basic phenomena and applications, as well as phenomena unique to small junctions. He also delved into fluctuation effects in conventional superconductors, high-temperature superconductors, and specialized topics like the Bogoliubov method, magnetic perturbations and gapless superconductivity, time-dependent Ginzburg–Landau theory, and nonequilibrium superconductivity.

The superconducting tunnel junction is a type of Josephson junction that takes advantage of quantum tunneling and superconductivity to create the Josephson effect [6]. It is an electronic device consisting of two superconductors, such as niobium, tantalum, or hafnium, separated by a very thin layer of insulating material, such as aluminium oxide. To fabricate a superconducting tunnel junction on an insulating substrate such as silicon, sapphire, or magnesium fluoride, the first layer of a thin film of superconducting metal with a typical thickness of several nanometers is deposited inside a vacuum chamber. This layer is then oxidized to form a high-quality insulating tunnel barrier after oxygen gas is introduced into the chamber. The final layer of superconducting metal is overlapped with the top of the previous layers when the vacuum is restored.

The superconducting tunnel junction has a wide range of practical applications [7,8], including high-sensitivity radio astronomy [9], single-photon detection [10–14], superconducting quantum interference devices (SQUIDS) [15,16], quantum computers [17], rapid single flux quantum (RSFQ) fast logic circuits, and Josephson voltage standards [18,19]. It is also used in high-sensitivity detectors of electromagnetic radiation, magnetometers, high-speed digital circuit elements, and

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quantum computing circuits [20,21]. Additionally, it finds important applications in precision measurements of voltages and magnetic fields, multijunction solar cells, quantum-mechanical circuits, superconducting qubits [22,23], phase-coherent caloritronics [24–27], as well as RSFQ digital electronics [28,29].

Axion detection is a cutting-edge particle detection technique with immense potential for advancing various scientific disciplines, particularly in relation to Josephson junctions. Detecting axions could yield crucial insights into the nature of dark matter and help unravel some of the greatest mysteries of modern cosmology. Several studies [30–34] have already been conducted in this field, exploring various approaches and experimental setups that utilize Josephson junctions for axion detection. These efforts have not only enhanced the technical capabilities of axion detectors but have also contributed to a deeper understanding of the fundamental physics involved.

However, superconducting tunnel junction devices are not isolated from their environment and are subject to fluctuations and interactions with surrounding electromagnetic fields, thermal noise, and other external factors [35]. Noise, including both classical sources such as Johnson-Nyquist noise and voltage fluctuations, is usually observed during experiments involving measurement operations. It is typically unwanted but inescapable due to factors such as thermal fluctuations, mechanical vibrations, industrial influences of voltage from power supplies, measurement operations [36]. Even though quantum noise is an uncontrolled variation deviating from a desired value, an accurate description of the superconducting tunnel junction device performance should seriously take into account these random effects [37]. Parisi's nobel lecture [38] integrated insights emphasizing noises role in systems with multiple equilibria, such as spin glasses and Josephson junction arrays. Stassi, De Liberato, Garziano, Spagnolo and Savasta [39] created multipartite nonclassical states, with the possibility to control the long-range quantum correlations of the emitted photons. They presented the way to new kinds of quantum fluids of light, arising from modulated vacuum fluctuations in linear systems. Carollo, Spagnolo and Valenti [40] derived a closed form expression for the symmetric logarithmic derivative of Fermionic Gaussian states, highlighting noise-induced coherence and quantum phase transitions. Magazzù, Valenti, Carollo and Spagnolo [41] compared Josephson junction switching dynamics with noise-driven transitions in biological ion channels. Spezia, Curcio, Fiasconaro, et al. [42] discussed noise-facilitated signal transduction in neural networks of biological systems. Agudov, Dubkov, Safonov, et al. [43] showed how noise stabilizes resistive switching in neuromorphic devices, and contrasted the adaptive biasing approach with noise-utilizing control in memristor systems, underscoring shared principles in leveraging fluctuations.

The resistively shunted superconducting tunnel junction system, in which the Josephson junction is connected in parallel with a shunting resistor, is an important building block of superconducting electronic circuits [44]. It is a fundamental component characterized by a critical current that flows through the junction without resistance. In spite of this, when a voltage bias is applied across the junction, the system enters a non-zero voltage state because of the tunneling of Cooper pairs from one superconductor to the other. The biased state is accompanied by the generation of heat, which in turn leads to the influence of thermal fluctuations on the system's dynamics [45,46]. The dynamics of a resistively shunted superconducting tunnel junction are influenced by the critical current and voltage state of the system.

Stochastic thermodynamics particularly refers to a system's behavior that is characterized by random fluctuations caused by the heating effect of thermal noise [47]. It has emerged as a powerful tool for understanding the performance of superconducting electronic circuits, including the role of noise and fluctuations in the behavior of these systems. In the resistively shunted superconducting tunnel junction system, these fluctuations result in a time-varying voltage and current across the junction, making its behavior unpredictable. The stochastic dynamics of this noise-affected system reveal that these fluctuations

could significantly impact the system's behavior [48]. Specifically, the thermal fluctuations could influence the switching behavior of the junction, affecting its transition between superconducting and biased states. Additionally, these fluctuations also affect the voltage-current characteristics of the junction, leading to deviations from the ideal behavior predicted by theoretical models. Understanding and modeling this stochastic behavior is crucial for optimizing the performance of superconducting electronic circuits [49]. This field of study has particular relevance for the design and operation of Josephson junction devices, which are highly sensitive to noise and fluctuations. In this context, stochastic thermodynamics can be used to guide the development of new strategies for minimizing the impact of noise on Josephson junction performance [50]. Moreover, Josephson junctions have been explored as potential noise detectors, leveraging their sensitivity to fluctuations in various physical quantities [51–53].

The studies of stochastic dynamics in the resistively shunted superconducting tunnel junction system in the presence of thermal noise constitute an active area of research. McCumber [54] demonstrated that the AC (Alternating Current) impedance observed by the junction has an impact on both the response time to changes in bias and the DC (Direct Current) voltage-current characteristics. Kautz [55] designed zero-bias Josephson voltage standards by utilizing superconducting tunnel junctions as the core component and calculated the quasipotential difference that represents the likelihood of escape from a basin of attraction for nonequilibrium systems at low temperatures. Spiechowicz, Hänggi and J. Łuczka [56] investigated the stochastic dynamics in an asymmetric SQUID composed of a loop containing three Josephson junctions, identified specific parameter sets where the ratchet effect is most pronounced, and showed how the direction of transport can be precisely controlled by adjusting the external magnetic flux. Guarcello, Bergeret and Citro [57] conducted an investigation into the switching current distributions exhibited by ferromagnetic anomalous Josephson junctions while subjecting them to a linearly increasing bias current. Their study analyzed the intricate interplay among noise, magnetization, phase dynamics, and the statistical characteristics of the switching current distribution. In addition to the studies of noise effects in conventional Josephson junctions, the impact of thermal fluctuations on anomalous Josephson junctions has also been explored, with recent work [58,59] in this area.

The primary aim of this research is to understand the stochastic dynamics of the specific resistively shunted superconducting tunnel junction system under the external excitation of thermal fluctuations. This knowledge can aid in the development of more accurate models for predicting the system's behavior, as well as exploring new techniques for controlling and mitigating the impact of thermal fluctuations on circuit performance.

This research investigates the stochastic dynamics of the resistively shunted superconducting tunnel junction system through phase-space analysis and computational methods. Specifically, it focuses on the system's response to various conditions, including the current parameter, damping coefficient, and thermal fluctuations. The discussion is further enhanced by deriving the Fokker-Planck equation for the resistively shunted superconducting tunnel junction system under thermal fluctuations. This equation provides a comprehensive framework for understanding the probabilistic evolution of the system's state over time, accounting for both deterministic and stochastic influences. Solving this equation yields deeper insights into the underlying mechanisms governing the system's behavior. Furthermore, numerical simulations assess the probability density of the system's state under specific parameter configurations. These simulations explore how the system's behavior changes as the current parameter, damping coefficient, and strength of thermal fluctuations are varied. This represents a significant advancement, as it provides more accurate predictions about the system's performance in real-world applications, ultimately contributing to the development of superconducting devices and technologies.

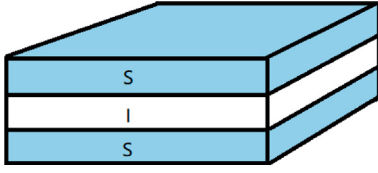


Fig. 1. A SIS junction consists of two bulk superconductors separated by a thin insulator layer.

Simulations reveal a previously unreported regime where intermediate thermal noise extends the lifetimes of metastable states, associated with the mean first passage time and noise enhanced stability. This counterintuitive result, stemming from barrier renormalization, offers new design principles for Josephson junction-based qubits.

This work is arranged as follows. In Section 2, the expression for the total junction current is explored according to Kirchhoff's current law, making it dimensionless. The fluctuation–dissipation theorem is then applied to introduce noise, thereby obtaining the resistively shunted superconducting tunnel junction system driven by thermal noise. In Section 3, it is found that the resistively shunted superconducting tunnel junction system corresponds to an idealized Hamiltonian system with resistance, current bias and noise intensity, which are introduced to model resistive losses, bias currents and thermal fluctuations. When the conservative Hamiltonian system does not include dissipative or externally driven terms, the energy surface and phase portrait are simulated. When the resistive damping and current bias parameters are non-zero, a clockwise hysteresis is observed in the typical I-V characteristic curve of the system, and a bifurcation of a planar limit cycle occurs. Most importantly, the stochastic dynamics of the system influenced by thermal fluctuation are depicted through numerical simulations. In Section 4, the Fokker–Planck equation for the resistively shunted superconducting tunnel junction system under the impact of thermal fluctuations is calculated, and the probability density is simulated based on this equation for certain parameter settings. In Section 5, the mean first passage time and noise enhanced stability are considered. In Section 6, the findings are summarized and several future threads of discussion are identified.

2. Dimensionless model

The seminal discussions of the Schrödinger equation in a classical context related to superconductivity were presented by Feynman, Leighton and Sands [60]. The Josephson equations of supercurrent and voltage were derived by employing the wave function of the superconducting tunnel junction and the time-dependent Schrödinger equation it satisfies. The notation and specific applications to Josephson junctions closely follow the comprehensive treatment provided by Barone and Paternò [3]. Landau and Lifschitz [61] also mentioned the alternative derivation with general argument that emphasizes macroscopic quantum phenomena in superconductors, and highlighted its relevance to both fundamental physics and emerging quantum technologies. Furthermore, Citro, Guarcello and Pagano [62] delved into the foundational principles of the Josephson effect, the remarkable properties of Josephson junctions, and the noise-driven switching dynamics of critical components in quantum computing and sensing. They emphasized the relevance of control strategies to mitigating decoherence in superconducting qubits.

Referring to Fig. 1, the basic Josephson junction (or SIS junction) is built from two bulk superconductors, separated by a very thin layer of an insulator. The separation layer is of the order of few angstroms to few nanometers.

When the superconductor's temperature T is cooled below its critical temperature T_c , the electrons within it begin to pair up, forming what are known as Cooper pairs. These electron pairs have a boson

nature, allowing all the electrons in the superconductor to condense into the ground state. This condensation allows the electrons in the superconductor to be treated as a single quantum particle, described by a macroscopic wave function that possesses a well-defined quantum phase:

$$\Psi = \Psi_0 e^{i\varphi}. \quad (2.1)$$

The wave functions of upper and lower superconductors are:

$$\begin{aligned} |U\rangle &= \sqrt{\rho_U} e^{i\varphi_U}, \\ |L\rangle &= \sqrt{\rho_L} e^{i\varphi_L}, \end{aligned} \quad (2.2)$$

where $\rho_U = \rho_L =: \rho > 0$ is assumed to be identical amplitude, which is the Cooper pair density in both upper and lower superconducting bulks. The total Hamiltonian $H := H_U + H_L + H_M$ consists of three terms including the Hamiltonians of the upper and lower superconducting bulks and a mix of them together:

$$\begin{aligned} H_U &= E_U |U\rangle\langle U|, \\ H_L &= E_L |L\rangle\langle L|, \\ H_M &= K(|U\rangle\langle L| + |L\rangle\langle U|), \end{aligned} \quad (2.3)$$

where $E_U = eV$ and $E_L = -eV$ are the energy levels in the two superconductors, and the constant K is a characteristic of the junction. Notice that the energy difference $E_U - E_L$ between the upper and lower superconductors is $2eV$, where V represents the voltage across the Josephson junction and e is the elementary charge (approximately 1.602×10^{-19} coulombs), and k_B is Boltzmann's constant (approximately 1.381×10^{-23} joules per kelvin). Other fundamental constants used in this work, such as Planck's constant h and the reduced Planck's constant $\hbar$, are defined for the sake of consistency and completeness.

The time-dependent Schrödinger equation $i\hbar\dot{\Psi} = H\Psi$ for the states $|U\rangle$ and $|L\rangle$ is:

$$\begin{cases} i\hbar \frac{d|U\rangle}{d\tau} = E_U |U\rangle + K |L\rangle, \\ i\hbar \frac{d|L\rangle}{d\tau} = E_L |L\rangle + K |U\rangle, \end{cases} \quad (2.4)$$

where $\hbar = h/2\pi$.

Noting that the evolution of Josephson phase is

$$\frac{d\phi}{d\tau} = \frac{d\varphi_L}{d\tau} - \frac{d\varphi_U}{d\tau} = -\frac{1}{\hbar}(E_L - E_U) = \frac{2eV}{\hbar}.$$

This is actually the time derivative of the quantum phase difference on the junction.

The two Josephson equations are

$$I = I_c \sin(\phi), \quad (2.5)$$

$$V = \frac{\hbar}{2e} \frac{d\phi}{d\tau}, \quad (2.6)$$

where I is the supercurrent through the junction, $I_c := \frac{2AK\rho}{h}$ is a junction parameter called the critical current for which A is a geometrical factor depended on the area of the junction, and V is the voltage across the junction. These equations establish a relationship between the supercurrent I and the voltage V with the phase ϕ serving as an intermediate variable.

We connect phase of the wave functions to current and voltage across weak link by the Eqs. (2.5)–(2.6). The Josephson equations indicate that a current I of magnitude less than I_c can flow without a voltage (DC Josephson effect), and that the current oscillates with a particular angular frequency $\omega_J := 2eV/\hbar$, called the Josephson frequency, when a constant voltage is present (AC Josephson effect).

According to Thévenin's theorem, the Josephson junction can be modeled as a parallel combination of a capacitor (representing junction capacitance) and a resistor (representing shunt resistance or quasiparticle tunneling). The superconducting closed loop configuration, such as in a SQUID, introduces additional flux quantization constraints, but this is not intrinsic to the Thévenin equivalence of a standalone junction. The AC impedance of the Josephson junction can be represented by a

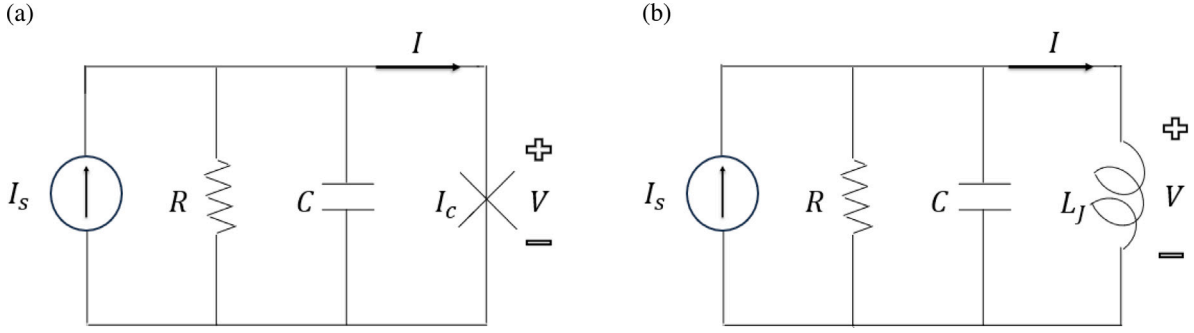


Fig. 2. (a) Full scheme of the experiment. The junction comprises three parallel lumped elements: a capacitor representing the internal capacitance of the junction, a shunt resistor, and a Josephson element having a current according to (2.5). The total junction current I_s is a standard bias current coming from a source in parallel to the Josephson junction; (b) When $\phi \ll 1$, the linear approximation $\sin(\phi) \approx \phi$ is valid, which means that $\sin(\phi)$ and ϕ are very close in this range. The system of motion, which consists of a single Josephson junction embedded in a superconducting loop, can be described by the equivalent circuit where the self inductance is L_J .

combination of a capacitor, a shunt resistor, and an ideal Josephson element, with all elements placed in parallel, as indicated in Fig. 2(a). Based on Ohm's Law in electrical circuit theory that describes the relationship between the current, voltage, and resistance, the normal current through the resistor is expressed mathematically as $\frac{V}{R}$, where R accounts for the resistance. The displacement current through the capacitor is $C \frac{dV}{d\tau}$, where C is the effective capacitance of the junction. In accordance with Kirchhoff's Current Law, the complete expression for the total junction current I_s is

$$I_s = C \frac{dV}{d\tau} + \frac{V}{R} + I = \frac{Ch}{2e} \frac{d^2\phi}{d\tau^2} + \frac{\hbar}{2eR} \frac{d\phi}{d\tau} + I_c \sin(\phi), \quad (2.7)$$

by utilizing the relation between the voltage and phase difference as given in Eq. (2.6). Eq. (2.7) describes the resistively and capacitively shunted junction model, which is a widely used model for the dynamics of Josephson junctions [54,63]. This model takes into account both the resistive and capacitive shunting effects that can occur in real-world Josephson devices, allowing for a more accurate description of their behavior.

In the context of the resistively and capacitively shunted junction model, the shunt resistance refers to an additional resistor placed in parallel with the junction. This resistor accounts for the dissipative processes that occur in the junction, such as the quasiparticle tunneling that occurs through the junction's gap [5]. The quasiparticle resistance $R_{SG}(T)$ (SG stands for subgap) is very large compared to the normal-state resistance R_N at low temperatures, but it decreases as the temperature increases and approaches R_N at the critical temperature T_c . The resistively and capacitively shunted junction model incorporates this shunt resistance in parallel with the junction's capacitive and inductive elements to describe the dynamics of the junction. The model is particularly useful for studying the switching behavior of Josephson junctions.

Further insight is gained from a mechanical analogue of the Josephson junction, which provides an intuitive understanding of its nonlinear dynamics. This analogue is based on the fact that the ideal Josephson element acts as an energy storage device. We make reasonable use of the Eqs. (2.5)–(2.6) to derive that the stored energy U , as a function of ϕ , is

$$U(\phi) = \int_0^\tau IV d\hat{\tau} = \int_0^\tau I_c \sin(\hat{\phi}) \frac{\hbar}{2e} \frac{d\hat{\phi}}{d\hat{\tau}} d\hat{\tau} = \frac{I_c \hbar}{2e} \int_{\phi_0}^\phi \sin(\hat{\phi}) d\hat{\phi} = E_J (1 - \cos(\phi)),$$

where $E_J := \frac{I_c \hbar}{2e}$ is the Josephson coupling energy, and the reference energy level is set to be at $\phi_0 = 0$. The system's nonlinearity arises from the sinusoidal nature of the potential, and the degree of nonlinearity is estimated by the Josephson coupling energy E_J , which determines the depth of the oscillations in potential energy.

If we interpret ϕ as a position coordinate, then the Eq. (2.7) can be interpreted as Newton's equation for the motion of a particle in the

potential $U(\phi)$ in the presence of damping and external applied forces. The acceleration of particle is as follows:

$$\frac{Ch}{2e} \frac{d^2\phi}{d\tau^2} = -I_c \sin(\phi) - \frac{\hbar}{2eR} \frac{d\phi}{d\tau} + I_s = -\frac{2e}{\hbar} U'(\phi) - \frac{\hbar}{2eR} \frac{d\phi}{d\tau} + I_s, \quad (2.8)$$

where

$$U'(\phi) = \frac{dU}{d\phi} = \frac{I_c \hbar}{2e} \sin(\phi)$$

is the derivative of the potential $U(\phi)$ with respect to ϕ . Therefore, the dynamical behavior of a Josephson junction is exactly analogous to the motion of a particle in a sinusoidal potential, often referred to as a washboard potential. The three elements on the right-hand side of Eq. (2.8) indicate the components contributing to the force. These components include the force resulting from the negative derivative $-U'(\phi)$ of the potential, a damping force that is proportional to the velocity $d\phi/d\tau$, and an externally applied force with a constant value. In the context of mechanics, the velocity of a particle represents the junction voltage, the viscous damping is equivalent to the junction conductance $G = \frac{1}{R}$, and the external force I_s represents the current bias [3]. Furthermore, the mass of the particle in the mechanical analogy is very similar to the junction capacitance on the left-hand side of Eq. (2.8).

When considering the case where the angle ϕ is much less than 1 (i.e., $\phi \ll 1$), we can approximate the sine function as $\sin(\phi) \approx \phi$. This approximation is linear in ϕ and can be extremely useful in simplifying calculations or analyzing the properties of functions that involve sine terms. By replacing the sine function with its linear approximation, we can often make complex expressions more manageable and gain valuable insights into the behavior of the system under consideration. Specifically in the case of $\phi \ll 1$,

$$I = I_c \sin(\phi) \xrightarrow{\sin(\phi) \approx \phi} I = I_c \phi \Rightarrow \frac{d\phi}{d\tau} = \frac{1}{I_c} \frac{dI}{d\tau} \xrightarrow{V = \frac{\hbar}{2e} \frac{d\phi}{d\tau}} V = \frac{\hbar}{2e I_c} \frac{dI}{d\tau},$$

where the ideal element $L_J := \frac{\hbar}{2e I_c}$ is the Josephson inductance of an inductor. It is related to the Josephson coupling energy by $E_J = L_J I_c^2$.

Due to the Eqs. (2.5)–(2.6), we apply the chain rule to calculate the time derivative of the supercurrent:

$$\frac{dI}{d\tau} = \frac{dI}{d\phi} \frac{d\phi}{d\tau} = I_c \cos(\phi) \frac{2e}{\hbar},$$

which can be rearranged in the form of the current–voltage characteristic of an inductor:

$$V = \frac{\hbar}{2e I_c \cos(\phi)} \frac{dI}{d\tau},$$

where

$$L(\phi) := \frac{\hbar}{2e I_c \cos(\phi)} = \frac{L_J}{\cos(\phi)}$$

provides the useful expression for the kinetic inductance as a function of the Josephson phase. It follows that $L(0) = L_J$. In essence, the kinetic behavior of the Josephson junction is similar to that of an inductor, stemming from the kinetic energy of the charge carriers.

In Fig. 2(b), the loop can be replaced by an equivalent circuit consisting of a capacitor and a shunt resistor, both parallel to the wire coil indicating symbolically inductance that is the tendency of an electrical conductor to oppose a change in the electric current flowing through it. The key idea is that the Josephson junction can be modeled as a nonlinear inductor, with its inductance varying as a function of the current. By replacing the Josephson junction with a capacitor and shunt resistor in parallel with a wire coil, it is suggesting that the Josephson junction can be approximated by this equivalent circuit. This approximation may be useful for simplifying the analysis or for understanding the behavior of the circuit under certain conditions.

Damping refers to the loss of energy from a system over time, often due to interactions with the environment. When a system is underdamped, it means that the system oscillates around its equilibrium position with a decreasing amplitude, but the oscillations do not die out completely. In contrast, an overdamped system would not oscillate at all, but would instead slowly approach its equilibrium position without any oscillations. Plasma oscillations are collective motions of charged particles in the potential well, and their dynamics can be described using the concept of damping.

The quality factor of the resonance of electrical circuits is a dimensionless parameter that characterizes the energy dissipation. It reflects the damping coefficient, which is given by $Q := R(\frac{C}{L_J})^{\frac{1}{2}} = R(\frac{2eI_c C}{\hbar})^{\frac{1}{2}}$. When Q exceeds $\frac{1}{2}$, the plasma resonance is underdamped. It means that the plasma oscillations are not completely damped out, but instead oscillate with a decreasing amplitude. Conversely, if Q falls below $\frac{1}{2}$, the plasma resonance is overdamped. The plasma frequency $\omega_p := \frac{Q}{RC} = (CL_J)^{-\frac{1}{2}} = (\frac{2eI_c}{Ch})^{\frac{1}{2}}$ of the junction is determined by the resonant frequency of RLC circuit. This can be an important characteristic, as it can affect the behavior of the system in various ways, such as its response to external perturbations or its ability to transfer energy.

After introducing the dimensionless scaling parameter ω_p , we rescale the time by it as $t = \omega_p \tau$. Thus Eq. (2.8) becomes

$$\frac{d^2\phi}{d\tau^2} = -\beta_J \frac{d\phi}{d\tau} - \sin(\phi) + \kappa,$$

where $\beta_J := \frac{1}{Q} = G(\frac{\hbar}{2eI_c C})^{\frac{1}{2}}$ is the dissipation coefficient, which stands for a parameter inversely related to the Josephson plasma frequency ω_p . Additionally, $\kappa := I_S/I_c$ represents the amplitude of the current bias normalized to the critical current I_c of the junction.

In Fig. 3(a), the circuit is shown at absolute zero temperature ($T = 0$), where thermal noise vanishes because it is proportional to temperature. Since the noise power spectral density is given by $4k_B T R$, thermal noise goes to zero at $T = 0$. Despite this, the resistor R itself still exists as a physical component within the circuit. It still has resistance R , which affects the circuit's behavior (e.g., damping or energy dissipation), even though there is no thermal noise. Moreover, quantum effects dominate at zero temperature. Even in the absence of thermal noise, quantum fluctuations (such as zero-point fluctuations) can still occur. These are not thermal in nature but arise from the Heisenberg uncertainty principle. On the other hand, Fig. 3(b) describes a situation at any finite temperature, where the resistor not only provides resistance but also introduces thermal noise $N(\tau)$ into the system. This noise is often modeled as a random current source in parallel with the noiseless resistor. The thermal noise in a resistor is a direct consequence of its dissipative nature. This representation is based on the fluctuation-dissipation theorem, which connects the fluctuations in a system's properties (such as its electrical conductivity) to its dissipative properties (such as its resistance to the flow of current). In summary, the presence or absence of thermal noise does not affect the

existence of the resistor as a component of the circuit. However, the behavior and representation of the resistor differ depending on whether thermal noise is present or absent. The current source $N(\tau)$ represents a white Gaussian noise process with zero mean and autocorrelation function

$$\langle N(\tau_1)N(\tau_2) \rangle = \frac{2k_B T}{R} \delta(\tau_1 - \tau_2), \quad (2.9)$$

where the Dirac delta function δ in time indicates that there is no correlation of the impacts between any distinct infinitesimal time increments $d\tau_1$ and $d\tau_2$ over a finite time interval. The thermal noise introduces uncertainty and can drive transitions between states. The thermal energy $k_B T$ sets the scale for the noise power, which is the product of Boltzmann constant k_B and temperature T . In terms of dimensionless quantities, Eq. (2.9) is equivalent to the dimensionless noise with a zero mean value and a second moment given by

$$\langle \zeta(t_1)\zeta(t_2) \rangle = 2\beta_J \Gamma \delta(t_1 - t_2),$$

where $\zeta = N/I_c$ is the normalized noise current, and

$$\Gamma := \frac{k_B T}{E_J} = \frac{\text{the thermal energy}}{\text{the Josephson coupling energy}}$$

corresponds to the dimensionless temperature. With this noise source added to the circuit, the equation of motion becomes

$$\frac{d^2\phi}{d\tau^2} + \beta_J \frac{d\phi}{d\tau} + \sin(\phi) = \kappa + \zeta(\tau). \quad (2.10)$$

3. Stochastic dynamics

Explicitly, we consider the resistively shunted superconducting tunnel junction driven by the thermal noise according to the stochastic model

$$\ddot{x} + \beta_J \dot{x} + \sin x = \kappa + \sqrt{2D}\eta(t), \quad (3.11)$$

where x represents a phase difference influenced by thermal fluctuation, $\kappa = I_S/I_c$ is related to the total current across the junction, $\beta_J = G(\frac{\hbar}{2eI_c C})^{\frac{1}{2}} = \frac{1}{Q}$ stands for a parameter inversely related to the Josephson plasma frequency $\omega_p = (CL_J)^{-\frac{1}{2}} = (\frac{2eI_c}{Ch})^{\frac{1}{2}}$, and

$$\frac{dx}{d\tau} = \omega_p \frac{dx}{dt}.$$

The overdot denotes differentiation with respect to time t . The interaction with thermal bath is modeled by δ -correlated, Gaussian white noise $\eta(t)$ of vanishing mean and unit intensity, i.e.,

$$\langle \eta(t) \rangle = 0, \quad \langle \eta(t)\eta(s) \rangle = \delta(t - s). \quad (3.12)$$

The processes $\eta(t)$ and $\frac{1}{\sqrt{2D}}\zeta(t)$ have the same distribution, where the noise intensity $D := \beta_J \Gamma$ is a measure of the strength of the fluctuating force. So the solution of stochastic system (3.11) coincides with that of system (2.10) in distribution.

The resistively shunted superconducting tunnel junction model in the absence of noise is

$$\ddot{x} + \beta_J \dot{x} + \sin x = \kappa. \quad (3.13)$$

We define a new variable $v := \dot{x}$. Then the second-order differential Eq. (3.13) can be transformed into a planar system of the form

$$\begin{cases} \dot{x} = v, \\ \dot{v} = \kappa - \beta_J v - \sin x, \end{cases} \quad (3.14)$$

where $\xi := \beta_J v$ is proportional to the voltage across the junction.

When $\kappa = \beta_J = 0$, the dynamical system (3.14) becomes a Hamiltonian system having one degree of freedom given by

$$H(x, v) = \frac{v^2}{2} - \cos x, \quad (\text{kinetic energy} + \text{potential energy})$$

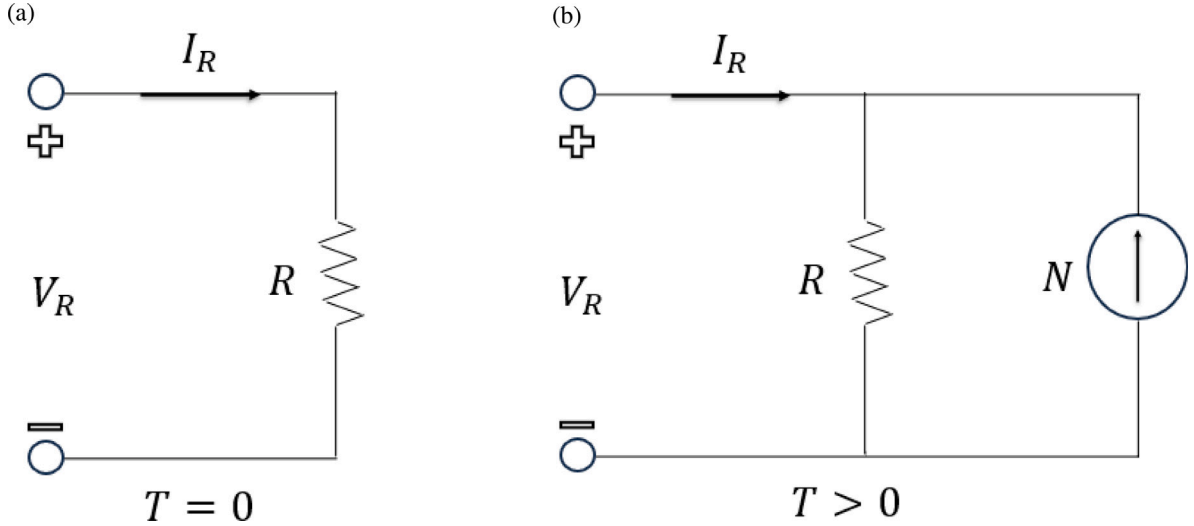


Fig. 3. (a) The resistor R exists when the temperature T of the circuit is set to zero, but there is no thermal noise because thermal noise depends on temperature; (b) At any non-zero temperature, the resistor's behavior is effectively modeled as a combination of a noiseless shunt resistor (pure resistance R), and a thermal noise source N in parallel with it, representing the random fluctuations due to temperature, as dictated by the fluctuation-dissipation theorem.

and Hamilton's equations are as follows:

$$\begin{cases} \dot{x} = v, \\ \dot{v} = -\sin x. \end{cases}$$

The equilibria of the above planar system are points where $\frac{\partial H}{\partial x} = \sin x = 0$ and $\frac{\partial H}{\partial v} = v = 0$. Thus the fixed points occur at $(n\pi, 0)$ in the (x, v) plane, where n is an integer. The Jacobian matrix is

$$J = \begin{pmatrix} 0 & 1 \\ -\cos x & 0 \end{pmatrix}.$$

For $m \in \mathbb{Z}$, the equilibria at $((2m+1)\pi, 0)$ are saddle points with eigenvalues $\lambda = \pm 1$ of J when $n = 2m+1$, while the equilibria at $(2m\pi, 0)$ are topological centers with eigenvalues $\lambda = \pm i$ of J when $n = 2m$. The fixed points are hyperbolic if n is odd and nonhyperbolic if n is even. Note that the Hartman-Grobman theorem [64] requires that the equilibrium must be hyperbolic. Therefore, Hartman-Grobman theorem cannot be applied when n is even.

Since the Hamiltonian energy is conserved, the motion is along contours of the energy H in the phase space (x, v) . Fig. 4(a) shows the energy diagram of a superconducting tunnel junction. Solution curves are sketched in Fig. 4(b) for which the axes are the displacement x and angular velocity v . The closed curves in Fig. 4(b) characterize local minima on the surface $H(x, v)$ as in Fig. 4(a), and the unstable fixed points correspond to local maxima on the same surface. The rotating curves surrounding the topological centers $(2m\pi, 0)$ represent bounded and periodic oscillations. We find that the heteroclinic connections at the saddle points $((2m+1)\pi, 0)$ form the separatrix. The wavy lines for large velocities describe librating motions.

For nonzero κ and β_J , interesting dynamics are observed. We investigate some of these via the sketch of typical current-voltage (I-V) characteristic curve of a superconducting tunnel junction, as depicted in Fig. 5(a) for $\beta_J = 0.6$. There is a clockwise hysteresis. The vertical axis is the current parameter κ of the junction. The horizontal axis corresponds to the normalized direct current voltage $\langle \xi \rangle = \beta_J \langle v \rangle$, where $\langle v \rangle$ is the average of the maximum and minimum values of v in the long-time region with $t \in [0, 200]$. We analyze the system's behavior.

The red curve reflects the current for decreasing voltage. As the voltage $\langle \xi \rangle$ is decreased down to the gap voltage, the relation $\kappa = \langle \xi \rangle$ still holds until a critical state at which $\kappa_c \approx 0.6965$ and $\langle \xi \rangle = 2\Delta/e$, where Δ is an energy gap of the superconductor and e is the electron charge. Since the bias voltage $e\langle \xi \rangle$ exceeds twice the quantized value of superconducting energy gap Δ , the quasiparticle current arises, i.e., $\kappa >$

κ_c . The red bar at $\kappa = \kappa_c$ represents the direct current Josephson effect. There is a bifurcation from oscillatory behavior back to the zero-voltage point $\langle \xi \rangle = 0$. A direct current Josephson supercurrent flows under $\langle \xi \rangle = 0$.

The green curve depicts the current for increasing voltage. When $\langle \xi \rangle = 0$, Josephson current I_S flows up to a maximum threshold value I_c , i.e., $\kappa < 1$. If $0 < \langle \xi \rangle < 1$, then $\kappa = 1$, i.e., $I_S = I_c$. As $\langle \xi \rangle$ is increased further, the identity $\kappa = \langle \xi \rangle$ is valid when the current bias I_S exceeds the critical current I_c , i.e., $\kappa > 1$. There is a bifurcation to an oscillating tunneling current.

The fixed points of the dynamical system (3.14) of first-order autonomous differential equations are found by solving the equations $\dot{x} = \dot{v} = 0$. Set $v = 0$. Then $\dot{x} = 0$. It follows that $\dot{v} = 0$ if and only if $\sin x = \kappa$. If the current bias is less than the critical current, i.e., $\kappa < 1$, two equilibria are calculated as $(x, v) = (x_e, 0)$ and $(\pi - x_e, 0)$, where $x_e = \sin^{-1} \kappa$, $0 < x_e < \pi/2$. The first solution is stable and represents the superconducting state, whereas the second one is an unstable saddle point. Taking initial values of $\sin x_0 = 0.1$ and $v_0 = 0.1$, we plot an evolution of a limit cycle bifurcation using $\beta_J = 1.2$ as the parameter κ increases from $\kappa = 0.1$ to $\kappa = 2$ by step 0.1, as demonstrated in Fig. 5(b). The plot illustrates how the system's state, represented by the coordinates $(\sin(x), v)$, changes over time as the system evolves according to the governing equation. The horizontal axis measures $\sin x$; the vertical axis represents voltage.

For fixed $\beta_J = 1.2$, we take three snapshots of the evolution for different values of κ to describe the bifurcations that occur in the model (3.14) since the limit cycle appears. Fig. 6 presents three panels, each showing the trajectories of the system under different dynamics. It is important to note that all three panels share the same initial state, allowing for a direct comparison of the system's behavior under the various dynamics. When $\kappa = 0.1$, the path goes from the starting point $(\sin x_0, v_0) = (0.1, 0.1)$ to the ending point $(\sin x, v) = (0.1, 0)$, as illustrated in Fig. 6(a). As can be seen from Fig. 6(b), the blue curve moves from the initial position $(\sin x_0, v_0) = (0.1, 0.1)$ to the final position $(\sin x, v) = (0.6965, 0)$ in the case of $\kappa = 0.6965$. Fig. 6(c) shows a limit cycle in a resistively shunted Josephson junction when $\kappa = 2$. The trajectory starting from the point $(\sin x_0, v_0) = (0.1, 0.1)$ reaches this limit cycle.

3.1. Effects of current parameter

After specifying the equivalent noise index $\sqrt{2D} = 0.5$ and the parameter value $\beta_J = 1.2$ in system (3.11), we reasonably use the Monte

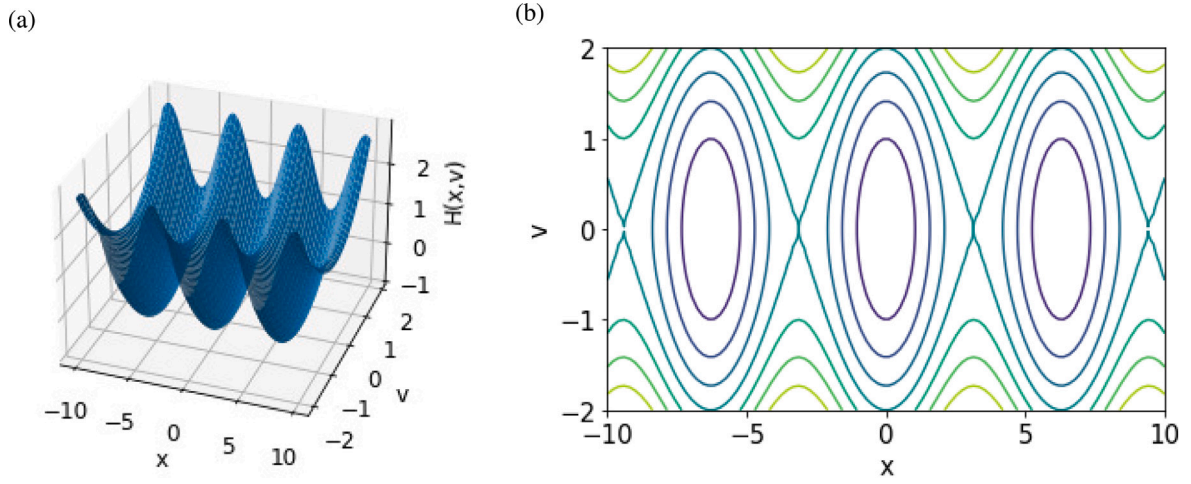


Fig. 4. (a) The energy surface $H(x, v)$; (b) A phase portrait of a Hamiltonian system is plotted in the (x, v) plane, where the vertical axis represents v , and the horizontal axis shows x .

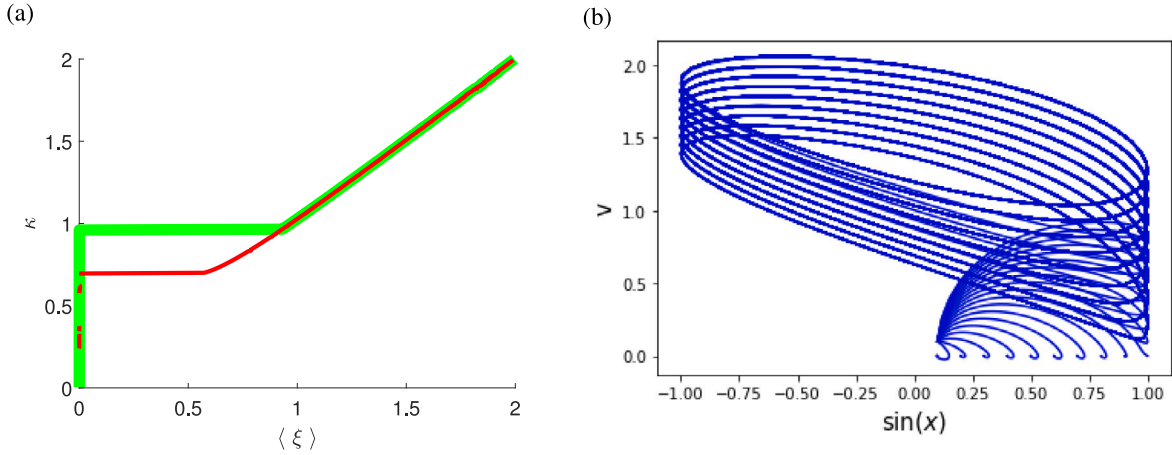


Fig. 5. (a) A typical I-V characteristic curve is simulated for a superconducting tunnel Josephson junction. It exhibits a clockwise hysteresis. The Cooper pair tunneling current appears when $\langle \xi \rangle = 0$, while the quasiparticle tunneling current is detected when $\langle \xi \rangle > 24/e$. Note that $\kappa = I_S/I_c$, where I_S represents the current through the Josephson junction, and I_c is a parameter of the junction known as the critical current; (b) An evolution of the bifurcating limit cycle of the system in the $(\sin(x), v)$ parameter space. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

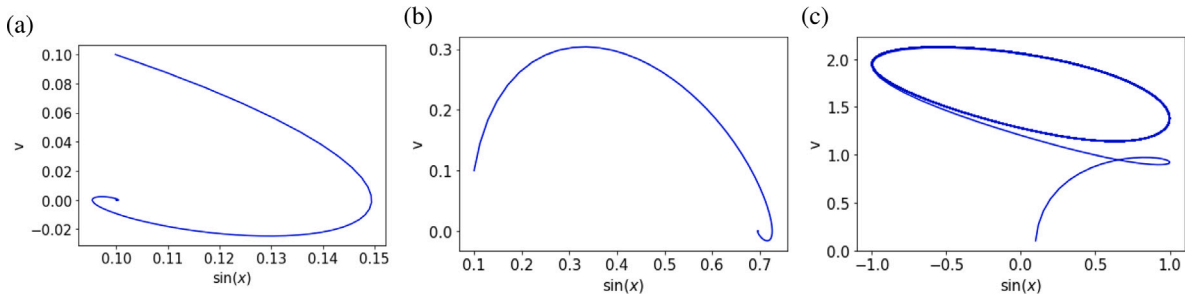


Fig. 6. Bifurcation of a planar limit cycle in a resistively shunted Josephson junction with $\beta_J = 1.2$: (a) $\kappa = 0.1$; (b) $\kappa = 0.6965$; (c) $\kappa = 2$.

Carlo method to perform detailed numerical simulations of dynamical orbits. Based on Figs. 7–9, we classify and identify transmission patterns in random media. In Fig. 7(a), considering the case where parameter $\kappa = 0.1$, the random disorder of the three-dimensional orbital evolution between time, displacement, and angular velocity is significantly strong. Fig. 7(b) exhibits periodic structures with a square-like symmetry, which shows a three-dimensional random block distribution of time, displacement, and angular velocity in a completely new form near the critical parameter $\kappa = 0.6965$. When $\kappa = 2$, the three-dimensional orbits between time, displacement, and angular velocity

in Fig. 7(c) are arranged along a diagonal random wall with varying spacing.

Fig. 7 reveals distinct patterns in the time evolution of the system's spatial configuration for the three different values of κ used. These patterns can be linked to the slope of the potential $U(\phi)$ in each case. For the smallest value of κ (Panel (a)), a lower bias current results in a smaller tilt (shallower slope) but a higher potential barrier, requiring more energy to escape. This makes escape harder, suppressing thermal/quantum escape and stabilizing the system in its metastable state. As κ approaches the critical value (Panel (b)), the tilt increases,

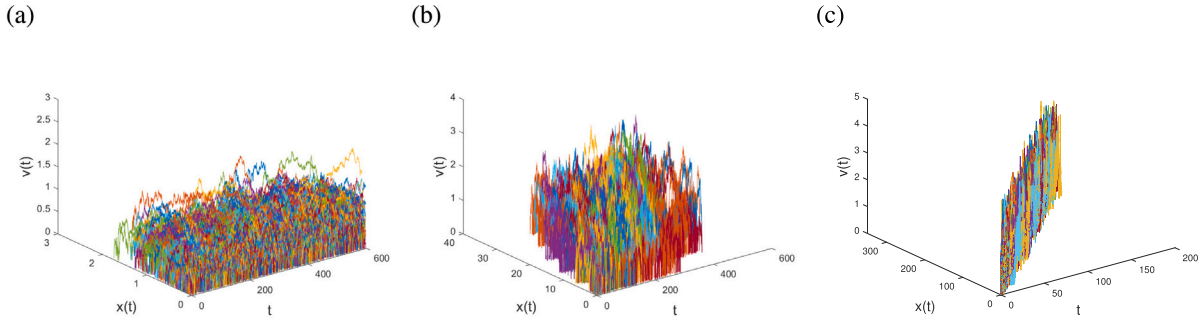


Fig. 7. The evolution of the variables $x(t)$ and $v(t)$ with respect to time characterized by system (3.11) for $\beta_f = 1.2$ and $\sqrt{2D} = 0.5$: (a) When $\kappa = 0.1$ and t ranges from 0 to 600, the variable x varies from 0 to 3, whereas in the stochastic sample trajectories, the variable v randomly decreases from 2 to 0; (b) When $\kappa = 0.6965$ and within the time range of 0 to 400, the value of $x(t)$ changes from 0 to 40, while the value of $v(t)$ fluctuates between 0 and 3. The resulting trajectories of this motion take on the shape of a fluctuating square; (c) When $\kappa = 2$ and t varies from 0 to 200, the time $x(t)$ evolves from 0 to 350, while concurrently, the function $v(t)$ oscillates rapidly within the range of 0 to 4. The functions $x(t)$ and $v(t)$ together form a wall exhibiting a random diagonal pattern during this time interval.

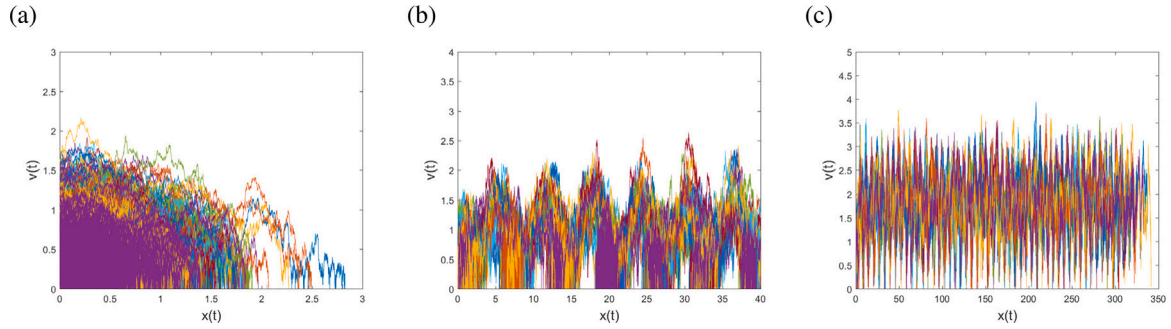


Fig. 8. The interaction between the variables $x(t)$ and $v(t)$ described by system (3.11) with $\beta_f = 1.2$ and $\sqrt{2D} = 0.5$: (a) When $\kappa = 0.1$, for most trajectories, $v(t)$ radiates from 2 to 0 when $x(t)$ reaches 2. However, for only a small fraction of trajectories, $v(t)$ radiates to 0 when $x(t)$ reaches 2.8; (b) When $\kappa = 0.6965$, $x(t)$ varies from 0 to 40, and $v(t)$ fluctuates within the range of 0 to 3. The resulting trajectories oscillate within a stochastic wave band; (c) When $\kappa = 2$, $x(t)$ evolves from 0 to 350, and $v(t)$ oscillates rapidly within the range of 0 to 4.

reducing the barrier height between minima and leading to faster switching. Therefore, increasing κ (closer to critical current) lowers the barrier, making escape easier. This means the system transitions to the voltage state (or next minimum) more quickly, which matches the expected dynamics. Finally, for the largest value of κ (Panel c)), it simultaneously reduces the potential barrier height, allowing the system to traverse the potential landscape more easily. This leads to even more pronounced oscillations. A larger κ tilts the potential more (steepens the slope), enhancing the escape rate and facilitating transitions to the voltage state. This is reflected in the smoother and more gradual changes observed in the system's motion, causing it to bounce back and forth more vigorously within the potential well. By linking the observed patterns in Fig. 7 to the slope of the potential $U(\phi)$ for the different values of κ , we gain a deeper understanding of how the shape of the potential landscape influences the system's dynamics.

Fig. 8 provides an alternative perspective on the system's dynamics by visualizing trajectory characteristics in a different parameter space. Fig. 8(a) reflects the dynamic behavior between displacement and angular velocity of the system at $\kappa = 0.1$. The trajectories do not follow a regular or predictable pattern. The values of $x(t) < \pi$ indicate that, within the chosen simulated timeframe, the noise amplitude is insufficient to enable the phase particle to escape the metastable state despite the small bias current κ (i.e., due to the high potential barrier). Consequently, the particle remains confined to oscillate within the potential orbit between displacement and angular velocity simulated in Fig. 8(b) at $\kappa = 0.6965$ exhibits a pattern of repeated fluctuations or oscillations with certain characteristics such as wavelength, amplitude, phase, and polarization. The oscillations display some degree of periodicity but are also influenced by stochastic factors. The behavior of the velocity $v(t)$ and a monotonic increase in the x -coordinate over time reflect

the intermediate bias current regime, where the reduced potential barrier enables the phase particle to overcome the barrier multiple times during the simulation. The peaks in $v(t)$ correspond to barrier crossings, while the minima signify re-trapping in adjacent potential wells. Such dynamics are indicative of the phase diffusion regime. In Fig. 8(c), the image between displacement and angular velocity in the stochastic superconducting system (3.11) shows the sharp nature of the trajectories, which appear to be closely spaced and aligned at $\kappa = 2$. In this regime, where $\kappa > 1$, i.e., the bias current exceeds the critical current I_c , the washboard potential becomes purely tilted (losing its metastable minima), causing the phase particle to roll freely down the potential. The particle rapidly attains a steady terminal velocity, determined by factors such as damping and noise intensity.

For an intuitive numerical simulation we plot the trajectories of the superconducting phase perturbed by thermal noise as a function of time. Unexpectedly, at $\kappa = 0.1$, the displacement trajectory exhibits multiple random fluctuations over time, as depicted in Fig. 9(a). These fluctuations resemble a complex and intertwining pattern that may suggest stochastic behavior within the system. The observed behavior in Fig. 9(a) at $\kappa = 0.1$, where the displacement trajectory appears multiplexed over time, is not necessarily unexpected given the system's initial state and the duration of the simulation. Specifically, the system starts at a minimum of the washboard potential and does not have sufficient time to escape and evolve by rolling down along the potential profile. The short duration of the simulation prevents switching events from occurring, which are expected to happen on average in a time that is approximately described by the Kramers formula [65]. Instead, Fig. 9(a) shows stochastic dynamics within the potential minimum at $\phi = \arcsin(\kappa)$.

If the parameter grows to $\kappa = 0.6965$, the displacement trajectory forms a random nonlinear microwave quantum network with wavelength switching, as suggested by Fig. 9(b). The displacement trajectory

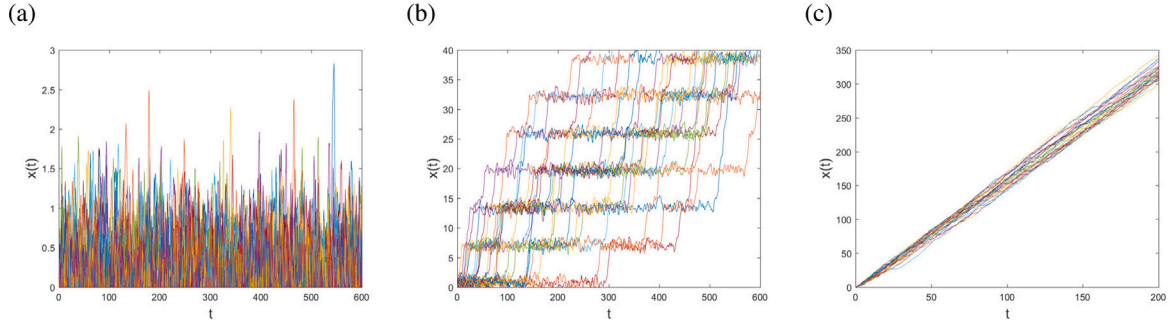


Fig. 9. In the stochastic superconducting system governed by Eq. (3.11) with parameters $\beta_J = 1.2$ and $\sqrt{2D} = 0.5$: (a) When $\kappa = 0.1$, the trajectories of the displacement $x(t)$ within the range $[0, 3]$ are multiplexed over the time range $t \in [600]$, generating a random pattern; (b) When $\kappa = 0.6965$, the trajectories of the displacement $x(t) \in [0, 40]$ form a tilted washboard pattern over the time range $t \in [600]$. Each of the seven steps within this pattern displays random amplitudes; (c) When $\kappa = 2$, the trajectories of the displacement $x(t) \in [0, 350]$ resemble a randomly fluctuating diagonal beam within the time range $t \in [200]$.

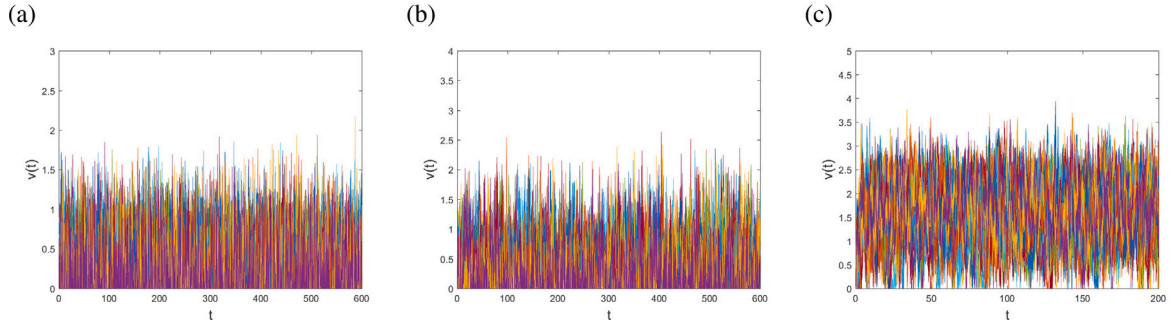


Fig. 10. In the stochastic superconducting system (3.11) with the specific parameter values of $\beta_J = 1.2$ and $\sqrt{2D} = 0.5$: (a) When $\kappa = 0.1$, the evolution of voltage $v(t)$ can reach a height of 2 within the time interval $t \in [0, 600]$. This evolution has a relatively dense concentration between 0 and 1, and a relatively sparse distribution between 1 and 2; (b) When $\kappa = 0.6965$, the voltage evolution $v(t)$ can attain a height of 2.5 over the time range $t \in [0, 600]$. The concentration is denser between 0 and 1.5, and becomes relatively sparse between 1.5 and 2.5; (c) When $\kappa = 2$, the voltage evolution $v(t)$ can attain a height of 4 within the time range $t \in [0, 200]$. During this period, the density of the voltage evolution is highest between 0.5 and 3, and becomes relatively sparse both below 0.5 and above 3, up to the height of 4.

in Fig. 9(b) exhibits stochastic jumps between the minima of the washboard potential, which are separated by 2π . These jumps occur with a stochastic frequency and are facilitated by the low height of the potential barrier at $\kappa \approx 0.7$. Given the imposed noise amplitude, the system is able to overcome the barrier and transition between the potential minima. Thus, the behavior shown in Fig. 9(b) represents stochastic dynamics within the washboard potential landscape.

Fig. 9(c) depicts the trajectory of the displacement in a stochastic superconducting system (3.11) with parameter $\kappa = 2$ changing randomly along a diagonal line, resulting in a large beam coverage range. Fig. 9(c) depicts the trajectory of the displacement in a stochastic superconducting system (3.11) with parameter $\kappa = 2$. As κ increases beyond 1, the washboard potential loses its characteristic maxima and minima structure, and the potential becomes more inclined. The figure shows the phase particle rolling on this slightly corrugated and tilted plane, where the effect of friction is also present.

For the parameters $\kappa = 0.1, 0.6965$ and 2, the voltage across the Josephson junction under external excitation presents different characteristics with time, we can observe the following phenomena in Fig. 10. As exhibited in Fig. 10(a), when $\kappa = 0.1$, the voltage evolution height can reach 2 over time, relatively dense between 0 and 1, and relatively sparse between 1 and 2. This indicates that the response of the Josephson junction is more concentrated in a smaller voltage range at lower parameters, and becomes relatively sparse as the voltage increases. When the parameter κ increases to 0.6965, the voltage evolution height over time reaches 2.5. It is concentrated between 0 and 1.5, while relatively sparse between 1.5 and 2.5, as rendered in Fig. 10(b). But The dense area of response is slightly larger than that in the previous case. This infers that the response of the Josephson junction becomes dense over a larger voltage range as the parameter κ increases. When $\kappa = 2$, the voltage evolution height over time can

reach 4. At this point, the response is dense between 0.5 and 3, while relatively sparse between 0 and 0.5 and between 3 and 4, as pictured in Fig. 10(c). Compared with the first two parameter values, when the parameter κ is further increased to 2, the dense region of the response expands to a larger voltage range, but it is concentrated in the middle voltage range.

The average speed of the phase particle after an escape phenomenon occurs is precise depending on bias current. The slope of the potential, friction, and thermal noise all play important roles in determining the particle's velocity. The more inclined the potential (i.e., the larger κ), the greater the average velocity at a fixed friction. These observations show that the voltage evolution of the Josephson junction under external excitation exhibits different distribution characteristics over time. These characteristics change as the parameter κ increases. At lower parameters, the response is concentrated in a smaller voltage range. However, with the parameters increasing, the response gradually expands to a larger voltage range while still maintaining a certain concentration area. The observed distribution may be influenced by both the intrinsic mechanism of the Josephson junction and the properties of external factors, such as thermal noise.

3.2. Effects of damping parameter

When β_J is less than 2, which corresponds to Q being greater than $1/2$, the plasma resonance is underdamped (or weakly damped), indicating a large capacitance and/or large resistance. The case where $\beta_J = 2$ is equivalent to Q being equal to $1/2$, representing a critical condition. Conversely, when β_J is greater than 2, which corresponds to Q being less than $1/2$, the plasma resonance is overdamped (or highly damped), suggesting a small capacitance and/or small resistance. After specifying the equivalent noise index $\sqrt{2D} = 0.5$ and the parameter

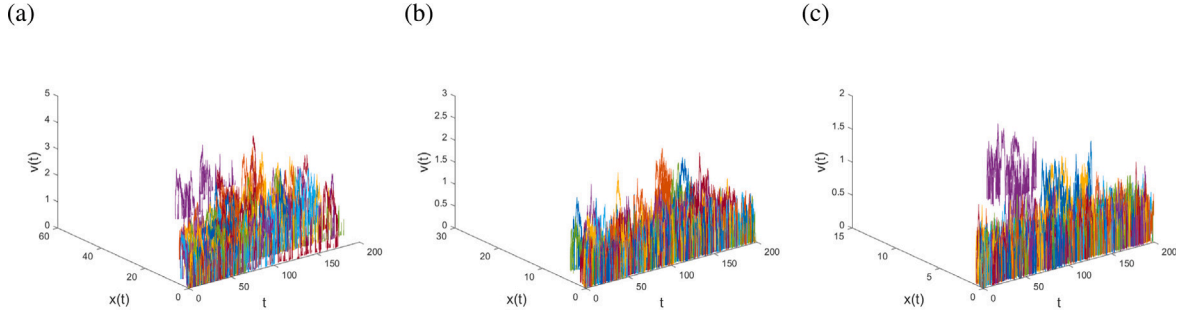


Fig. 11. In the stochastic superconducting system described by Eq. (3.11) with specific parameter values of $\kappa = 0.6965$ and $\sqrt{2D} = 0.5$, and given a time range $t \in [0, 200]$, we consider the following cases: (a) The phase and voltage oscillation ranges are $[0, 60] \times [0, 3]$ when $\beta_J = 1$; (b) The phase and voltage oscillation ranges are $[0, 22] \times [0, 2]$ when $\beta_J = 2$; (c) The phase and voltage oscillation ranges are $[0, 15] \times [0, 1.8]$ when $\beta_J = 3$.

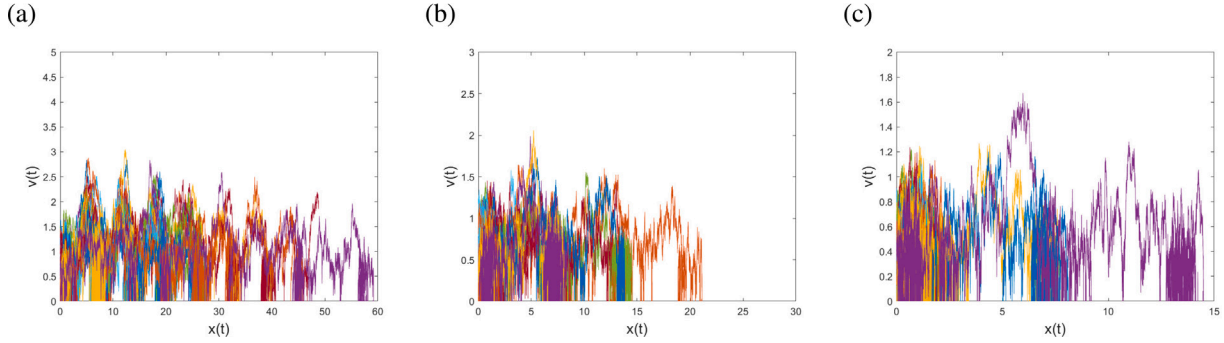


Fig. 12. For the stochastic superconducting system (3.11) with specific parameter values of $\kappa = 0.6965$ and $\sqrt{2D} = 0.5$, we observe the following: (a) In the underdamped case ($\beta_J = 1$), the phase $x(t)$ and voltage $v(t)$ form a random oscillatory wave with 9 peaks, exhibiting a downward trend. This indicates that the phase $x(t)$ evolves from 0 to 60, and the voltage $v(t)$ decreases toward 0; (b) In the critical case ($\beta_J = 2$), the phase $x(t)$ and voltage $v(t)$ oscillate with much shorter wavelengths and lower amplitudes compared to the underdamped case, with only 4 peaks; (c) In the overdamped case ($\beta_J = 3$), the phase $x(t)$ and voltage $v(t)$ oscillate with three peaks, and the majority of the oscillation is concentrated at the first peak.

value $\kappa = 0.6965$ related to the total current across the junction, we accurately describe the behavior of phase $x(t)$ and voltage $v(t)$ over time t in the stochastic superconducting system described by Eq. (3.11), within specific ranges, for different damping coefficients β_J . In the underdamped system, the most pronounced oscillations are observed, while in the overdamped system, the least pronounced oscillations are seen. The critical case lies between these two extremes.

Over the time interval $t \in [0, 200]$, the phase $x(t)$ and voltage $v(t)$ oscillate randomly within the ranges of $[0, 60] \times [0, 3]$ when $\beta_J = 1$ (underdamped). This is illustrated in Fig. 11(a). For the critical case $\beta_J = 2$, the phase $x(t)$ and voltage $v(t)$ oscillate randomly within the ranges of $[0, 22] \times [0, 2]$. This is shown in Fig. 11(b). When $\beta_J = 3$ (overdamped), the phase $x(t)$ and voltage $v(t)$ oscillate randomly within the range $[0, 15] \times [0, 1.8]$. This is depicted in Fig. 11(c).

When $\beta_J = 1$, the phase $x(t)$ and voltage $v(t)$ form a random oscillatory wave in the ranges of $[0, 60] \times [0, 3]$. There are 9 peaks in the random oscillation, and the peaks exhibit a downward trend, meaning that the phase $x(t)$ evolves from 0 to 60 while the voltage $v(t)$ decreases toward 0. The oscillations continue to decay but overshoot the equilibrium position, leading to a longer-lasting and more pronounced wave pattern, as displayed in Fig. 12(a). In contrast, the overdamped and critically damped systems decay more quickly and have shorter, less pronounced peaks. For the critical case $\beta_J = 2$, the phase $x(t)$ and voltage $v(t)$ oscillate randomly within the region $[0, 22] \times [0, 2]$, with much shorter wavelengths and lower wave heights, featuring only 4 peaks in the wave pattern, as represented in Fig. 12(b). The wavelengths and wave heights are significantly shorter and lower when compared to the underdamped scenario. For $\beta_J = 3$, the phase $x(t)$ and voltage $v(t)$ oscillate randomly within the range $[0, 15] \times [0, 1.8]$, with three distinct peaks in the wave pattern. The main oscillation or the most significant deviation from the baseline occurs at the first peak, as visualized in Fig. 12(c).

When $\beta_J = 1$, the phase $x(t)$ moves in a 9-layer tilted potential, with the motion mainly concentrated in the first 5 layers, as depicted in Fig. 13(a). The number of layers suggests differences in the frequency or damping rate of the oscillations. A higher number of layers in a similar time frame typically indicates a higher frequency or less damping. For the critical case $\beta_J = 2$, the phase $x(t)$ moves in a 4-layer tilted potential, with the movement mainly concentrated in the first and second layers, as shown in Fig. 13(b). When $\beta_J = 3$, the phase $x(t)$ moves in a 3-layer tilted potential, with the motion mainly concentrated in the first layer, as illustrated in Fig. 13(c).

Over the time interval $t \in [0, 200]$, the voltage $v(t)$ rapidly oscillates within the range $[0, 3]$ when $\beta_J = 1$, as visualized in Fig. 14(a). When $\beta_J = 2$, the voltage $v(t)$ rapidly oscillates within the range $[0, 2]$, as represented in Fig. 14(b). When $\beta_J = 3$, the voltage $v(t)$ rapidly oscillates within the range $[0, 1.8]$, as displayed in Fig. 14(c).

3.3. Effects of thermal noise

The behavior of the phase and voltage in the given system denoted as (3.11) is significantly influenced by the noise intensity $\sqrt{2D}$. This noise intensity affects not only the amplitude and frequency of the oscillations but also the shape of the potential landscape within which the phase evolves. Let us explore the effects of varying $\sqrt{2D}$ with fixed parameters $\kappa = 0.6965$ and $\beta_J = 2$.

Over the time interval $t \in [0, 200]$, the phase $x(t)$ and voltage $v(t)$ evolve along arc-shaped trajectories within the region $[0, 2.2] \times [0, 1]$ during switching dynamics with $\sqrt{2D} = 0.3$. This behavior arises from the interplay between Josephson oscillations and resistive damping under a moderate current bias $\kappa = 0.6965$. As illustrated in Fig. 15(a), the system (3.11) exhibits transient oscillations before settling into a metastable state, tracing semicircular paths. These trajectories suggest

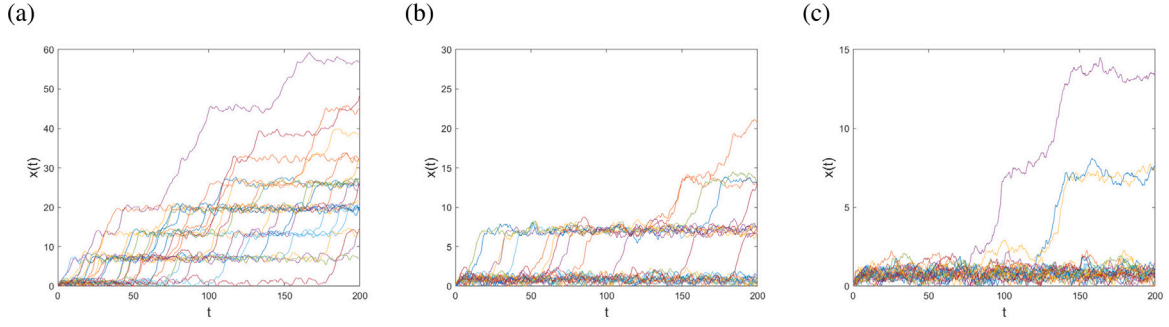


Fig. 13. In the stochastic superconducting system described by Eq. (3.11) with specific parameter values of $\kappa = 0.6965$ and $\sqrt{2D} = 0.5$, we observe the following: (a) When $\beta_J = 1$, the motion of $x(t)$ is mainly concentrated within the first 5 layers of a 9-layer tilted potential; (b) When $\beta_J = 2$, the motion of $x(t)$ is mainly concentrated within the first and second layers of a 4-layer tilted potential; (c) When $\beta_J = 3$, the motion of $x(t)$ is concentrated within the first layer of a 3-layer tilted potential.

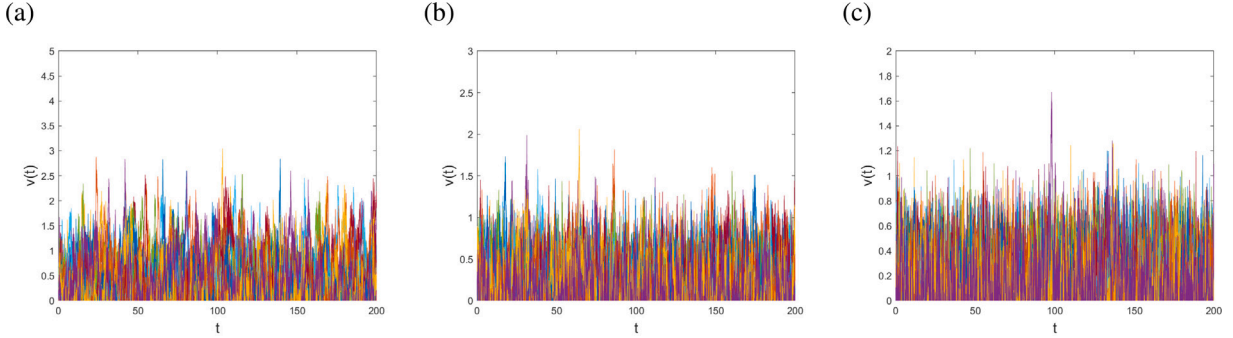


Fig. 14. In the stochastic superconducting system (3.11) with specific parameter values of $\kappa = 0.6965$, $\sqrt{2D} = 0.5$, we observe the following: (a) When $\beta_J = 1$, the voltage $v(t)$ rapidly oscillates within the range of 0 to 3. This represents that the underdamped system exhibits the most pronounced oscillations; (b) When $\beta_J = 2$, the voltage $v(t)$ rapidly oscillates within the range of 0 to 2. The critical case lies between the underdamped and overdamped systems; (c) When $\beta_J = 3$, the voltage $v(t)$ rapidly oscillates within the range of 0 to 1.8. The overdamped system shows the least pronounced oscillations.

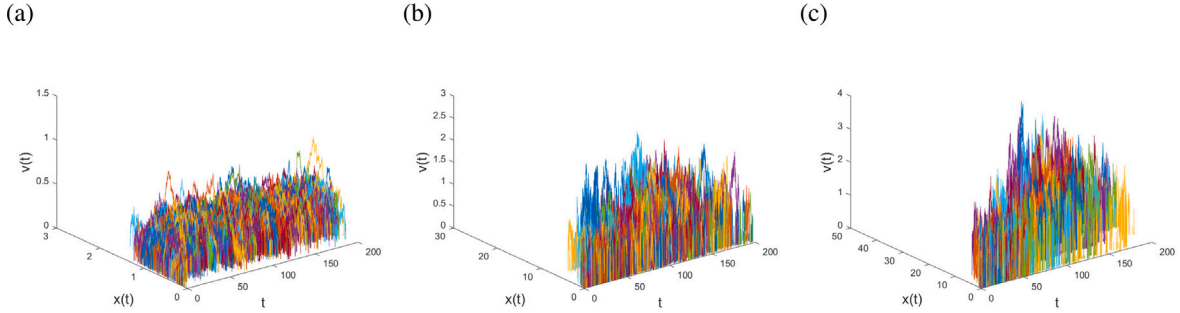


Fig. 15. The combined phase-voltage evolution over the time interval $t \in [0, 200]$ in the stochastic superconducting system (3.11), with specific parameter value of $\kappa = 0.6965$ and damping coefficient $\beta_J = 2$, is shown for different noise intensities: (a) $\sqrt{2D} = 0.3$; (b) $\sqrt{2D} = 0.6$; (c) $\sqrt{2D} = 0.9$.

relatively constrained oscillatory motion. When $\sqrt{2D} = 0.6$, the phase $x(t)$ and voltage $v(t)$ exhibit more extensive random vibrations within the region $[0, 26] \times [0, 2]$. This suggests a broader range of motion and increased noise, as shown in Fig. 15(b). When $\sqrt{2D} = 0.9$, the phase $x(t)$ and voltage $v(t)$ randomly vibrate in the extensive region $[0, 46] \times [0, 3]$, indicating a highly dynamic and noisy system, as depicted in Fig. 15(c).

When $\sqrt{2D} = 0.3$, the phase $x \in [0, 2.2]$, the voltage $v(t)$ randomly rises from 0 to 1 and then randomly falls back to 0, indicating a periodic but noisy behavior within a limited range, as displayed in Fig. 16(a). When $\sqrt{2D} = 0.6$, the phase $x \in [0, 26]$, the voltage $v(t)$ oscillates randomly within $[0, 2]$, as represented in Fig. 16(b). When $\sqrt{2D} = 0.9$ and the phase $x \in [0, 46]$, the voltage $v(t) \in [0, 3]$ oscillates randomly, as visualized in Fig. 16(c).

When $\sqrt{2D} = 0.3$, the phase $x(t)$ initially rises rapidly, and soon after, it oscillates around $x = 0.8$, indicating a stable equilibrium point with minor fluctuations, as depicted in Fig. 17(a). When $\sqrt{2D} = 0.6$, $t \in [0, 200]$, the phase $x(t)$ undergoes a tilted phase transition at 5

layers, suggesting a more complex transition dynamics due to increased noise, as shown in Fig. 17(b). When $\sqrt{2D} = 0.9$, $t \in [0, 200]$, the phase $x(t)$ undergoes a tilted phase transition at 8 layers, indicating an even more complex and layered transition dynamics compared to lower noise intensities, as illustrated in Fig. 17(c). At critical tilting, the phase particle escapes from a metastable potential minimum, marking a noise-driven transition to a resistive state. This refers to a multilayer junction structure (e.g., superconducting-insulator stacks) where the number of layers modifies the effective critical current, shifting the tilt required for phase escape.

When $\sqrt{2D} = 0.3$, the voltage $v(t) \in [0, 1]$ oscillates rapidly, as visualized in Fig. 18(a). As represented in Fig. 18(b), the voltage $v(t) \in [0, 2]$ oscillates rapidly when $\sqrt{2D} = 0.6$, indicating a more pronounced and widespread fluctuation compared to $\sqrt{2D} = 0.3$. When $\sqrt{2D} = 0.9$, the voltage $v(t)$ oscillates rapidly within $[0, 3]$, showing the largest range of motion and highest level of noise among the three cases, as displayed in Fig. 18(c). The oscillation height of voltage $v(t)$ increases

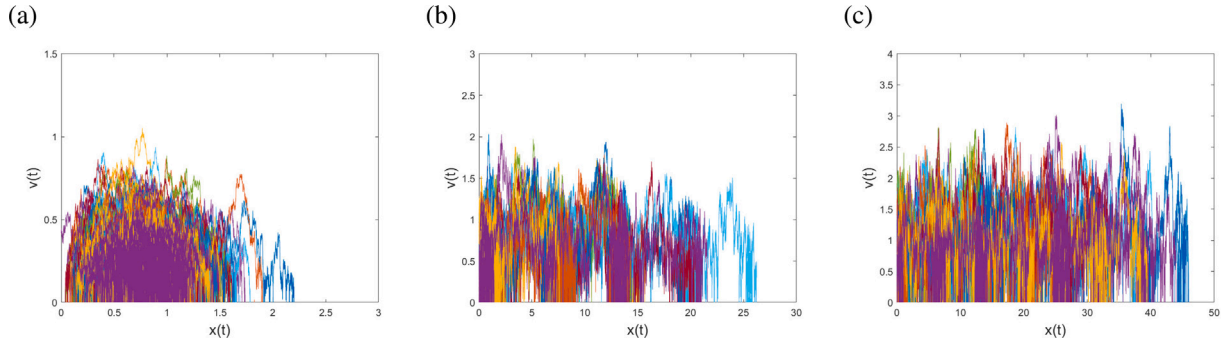


Fig. 16. The relationship between phase and voltage in the stochastic superconducting system (3.11), with specific parameter values of $\kappa = 0.6965$ and $\beta_J = 2$, demonstrates the increase in amplitude and range for each noise level with $\sqrt{2D}$: (a) For $\sqrt{2D} = 0.3$, the phase $x(t)$ and voltage $v(t)$ follow nonlinear trajectories in the phase-voltage plane, characterized by arc-shaped oscillatory profiles within the region $[0, 2.2] \times [0, 1]$. During these oscillations, the voltage $v(t)$ stochastically rises from 0 to 1 before falling back to 0; (b) For $\sqrt{2D} = 0.6$, the phase $x(t)$ and voltage $v(t)$ randomly vibrate within the range of $[0, 26] \times [0, 2]$, showing a downward trend; (c) For $\sqrt{2D} = 0.9$, the phase $x(t)$ and voltage $v(t)$ randomly vibrate within the region $[0, 46] \times [0, 3]$.

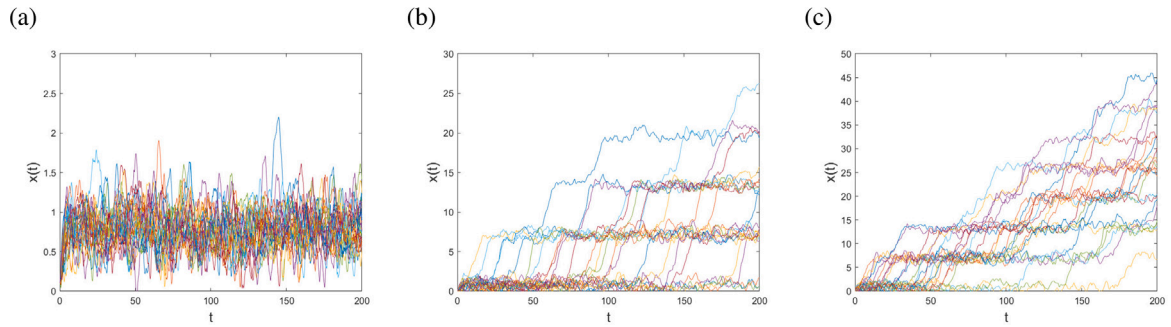


Fig. 17. The phase transition dynamics in the stochastic superconducting system (3.11), with specific parameter values of $\kappa = 0.6965$ and $\beta_J = 2$, indicate increasing complexity with higher noise levels: (a) The phase $x(t)$ initially rises rapidly and then oscillates around $x = 0.8$ when $\sqrt{2D} = 0.3$; (b) The phase $x(t)$ undergoes a tilted phase transition at 5 layers when $\sqrt{2D} = 0.6$; (c) The phase $x(t)$ is undergoing a tilted phase transition at 8 layers when $\sqrt{2D} = 0.9$.

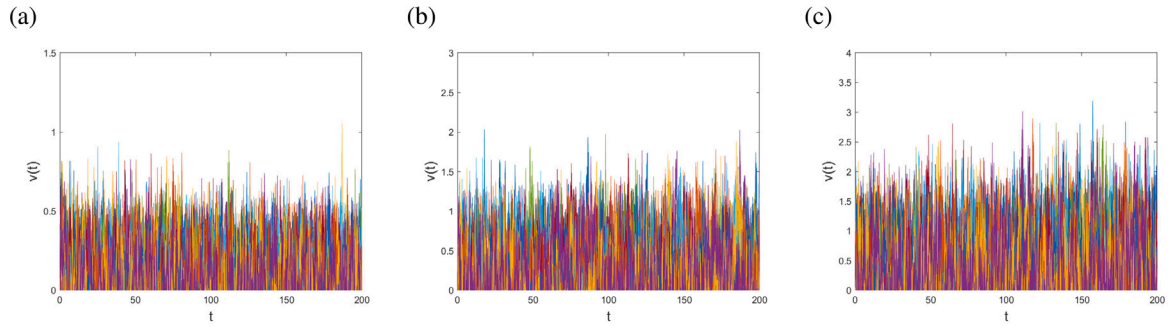


Fig. 18. The voltage oscillation over time in the stochastic superconducting system (3.11), with specific parameter values of $\kappa = 0.6965$ and $\beta_J = 2$, highlights varying ranges and randomness of voltage changes depending on the noise level $\sqrt{2D}$: (a) When $\sqrt{2D} = 0.3$, the voltage $v(t)$ exhibits rapid oscillation within $[0, 1]$; (b) When $\sqrt{2D} = 0.6$, the voltage $v(t)$ oscillates rapidly within the range of $[0, 2]$; (c) When $\sqrt{2D} = 0.9$, the voltage $v(t)$ oscillates rapidly within the range of $[0, 3]$.

with the increase of $\sqrt{2D}$. As $\sqrt{2D}$ increases from 0.3 to 0.9, the range of voltage oscillation increases from $[0, 1]$ to $[0, 3]$. This confirms that the noise intensity $\sqrt{2D}$ directly influences the amplitude of voltage oscillations, with higher noise leading to more extensive and dynamic voltage fluctuations.

In conclusion, the noise intensity $\sqrt{2D}$ plays a crucial role in modulating the behavior of the phase and voltage in the system (3.11), affecting both the amplitude and frequency of oscillations as well as the overall dynamics of the phase transitions.

4. Fokker-Planck equation

The Fokker-Planck equation is a partial differential equation that describes the time evolution of the probability distribution function of

a stochastic process. It is widely used in the study of systems subject to random fluctuations and thermal noise [3].

The mapping defined by Eq. (3.11) can also be thought of as two-dimensional. Rewrite the second-order stochastic differential equation (3.11) into two-dimensional first-order stochastic differential equations

$$\begin{cases} dx = v dt, \\ dv = (\kappa - \beta_J v - \sin x) dt + \sqrt{2D} d\eta(t), \end{cases} \quad (4.15)$$

where the thermal noise voltage source is Brownian motion. The state of the above system is specified by two variables x and $v = dx/dt$. At finite temperatures, these variables evolve according to a two-dimensional continuous Markov process, influenced by Brownian noise.

The equation of motion in abbreviated form is

$$\frac{d\mathbf{u}}{dt} = \mathbf{F}(\mathbf{u}) + \Sigma\Theta(t), \quad (4.16)$$

where the quantities are

$$\mathbf{u} = \begin{pmatrix} u \\ v \end{pmatrix}, \quad \mathbf{F} = \begin{pmatrix} v \\ \kappa - \beta_J v - \sin x \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{2D} \end{pmatrix},$$

$$\Theta(t) = \begin{pmatrix} 0 \\ \eta(t) \end{pmatrix}.$$

Within the framework of statistical mechanics, the time evolution is characterized predominantly by the phase space distribution function $p(\mathbf{u}, t)$ that determines the average properties of the system being in a specific state $\mathbf{u}$ at a particular time t . The probability of finding the system state within the region $d\mathbf{u}$ surrounding the point $\mathbf{u}$ at time t is given by $p(\mathbf{u}, t)d\mathbf{u}$. Rather than seeking a general solution to Eq. (4.16), we are interested in the probability density $p(\mathbf{u}, t)$ for the value of $\mathbf{u}$ at time t . Furthermore, what we really want is the average of this probability distribution over the noise. One way to find the noise average is to recognize that $p(\mathbf{u}, t)$ is a conserved quantity, ensuring that the total probability of describing the system somewhere is unity at all times. Mathematically, this is expressed as an integral over all possible states $\mathbf{u}$ of the phase space distribution function $p(\mathbf{u}, t)d\mathbf{u}$, which must add up to 1 at any given time. It means that

$$\int p(\mathbf{u}, t)d\mathbf{u} = 1, \quad \forall t.$$

The above conservation law is a fundamental principle in probability theory, which is crucial for maintaining that the description of the system's behavior is consistent and complete, making it a valuable tool in statistical mechanics and related fields. We expect that the time derivative of the conserved density $p(\mathbf{u}, t)$ is balanced by the divergence of a flux, which is the product of a velocity and that density. This is how the Liouville equation

$$\frac{\partial p}{\partial t} + \frac{\partial}{\partial \mathbf{u}} \left(\frac{d\mathbf{u}}{dt} p \right) = 0,$$

is derived in statistical mechanics. Here, the space coordinate is $\mathbf{u}$, the density at $\mathbf{u}$ is $p(\mathbf{u}, t)$, and the velocity at $\mathbf{u}$ is $d\mathbf{u}/dt$. When the time derivative $d\mathbf{u}/dt$ is replaced with the right-hand side of (4.16), we get

$$\frac{\partial p(\mathbf{u}, t)}{\partial t} = - \frac{\partial}{\partial \mathbf{u}} (\mathbf{F}(\mathbf{u})p(\mathbf{u}, t) + \Sigma\Theta(t)p(\mathbf{u}, t)).$$

Therefore, the Fokker–Planck equation describing the probability density $p(x, v, t)$ for the phase-space distribution is

$$\begin{aligned} \frac{\partial p(x, v, t)}{\partial t} &= \beta_J p(x, v, t) - v \frac{\partial p(x, v, t)}{\partial x} \\ &+ (\beta_J v + \sin x - \kappa) \frac{\partial p(x, v, t)}{\partial v} + D \frac{\partial^2 p(x, v, t)}{\partial v^2}. \end{aligned} \quad (4.17)$$

By virtue of Itô's formula and integration by parts, more accurate calculations for obtaining the Fokker–Planck equation can be found in the Refs. [66,67].

Having analyzed the evolution of $x(t)$ and $v(t)$ in the previous figures, we now turn our attention to Fig. 19, which depicts the system's kinetic energy over time. A striking observation is the direct correlation between the kinetic energy and the velocity profile seen in the $v(t)$ graph. During intervals of acceleration, where the velocity curve steepens, we see a corresponding surge in kinetic energy in Fig. 19. Conversely, during deceleration phases, when the velocity tapers off, the kinetic energy gradually diminishes. This alignment underscores the fundamental relationship between velocity and kinetic energy, reinforcing our understanding of the system's dynamic behavior. Furthermore, by combining the insights from all three graphs, we gain a deeper appreciation of the system's energy conversion mechanisms and the factors that govern its acceleration and deceleration.

For analytical treatments it is much more convenient to work with phase-space probabilities rather than individual stochastic trajectories.

As the point (x, v) evolves in the phase space, the three-dimensional simulations reveal a single peak in the distribution of $p(x, v, t)$ for a fixed time $t = 10$, as shown in Fig. 19. When the small parameter κ reaches 0.1, Fig. 19(a) shows that the peak value of the probability density function reaches 11. At the critical value $\kappa = 0.6965$, the peak of probability density actually decreases to 0.7, as clear from Fig. 19(b). This reduction contrasts the density profiles before and near criticality. In the case of $\kappa = 2$, the peak's height further decreases to 0.06, as exemplified in Fig. 19(c). Overall, the peak of probability density is inversely proportional to the parameter κ . This fact is corroborated by the comparative plots of the density profiles. At lower values of κ (high barrier), the system spends most of its time near the metastable minimum, resulting in a sharp peak in $p(x, v, t)$. As κ increases (lower potential barrier), escape events become more frequent, spreading the probability density across the free-running state, leading to a broader distribution of x and v (wider peak), and reducing the height of the probability peak because the system spends less time trapped in a single state.

Fig. 19 shows the time-dependent probability density $p(x, v, t)$, which directly relates to the dynamics of $x(t)$ and $v(t)$ in Figs. 8–9. For small κ (Fig. 19(a)), the sharp peak in $p(x, v, t)$ reflects the confinement of the phase particle to a single metastable well (Fig. 8(a)), with minimal exploration of phase space. For $\kappa = 0.1$, the $x(t)$ oscillations (Fig. 9(a)) correspond to a localized $p(x, v, t)$. As κ increases (Fig. 19(b)), the peak broadens and splits, corresponding to stochastic phase slips between wells (Fig. 8(b)), where $x(t)$ exhibits cumulative jumps and $v(t)$ shows intermittent spikes. For $\kappa = 0.6965$, intermittent escapes (Fig. 9(b)) broaden $p(x, v, t)$ into a broader distribution. For $\kappa = 2$ (Fig. 19(c)), the potential tilt eliminates metastable minima, resulting in a broad $p(x, v, t)$ centered at the terminal velocity, consistent with the free-running $x(t)$ in Fig. 8(c). The decay of the peak and the increase in width with κ thus quantify the transition from localized thermal fluctuations to noise-driven diffusion and deterministic sliding. For $\kappa = 2$ free-running $x(t)$ (Fig. 9(c)) leads to a broad, unimodal $p(x, v, t)$ centered at the terminal velocity. Higher values of κ allow the phase particle to explore a broader range of x and v due to increased noise-driven transitions (phase slips) or free-running motion. This reflects greater uncertainty (entropy) in the system's state, consistent with the phase diffusion regime (Fig. 8(b)) or resistive state (Fig. 8(c)).

Fig. 20(a) means that, under given conditions $\kappa = 0.6965$ and $\sqrt{2D} = 0.5$, when the damping parameter β_J is set to 1, the observed peak value of the probability density function (i.e., the maximum value of the function $p(x, v, t)$) lies within the interval of 10 000 to 15 000. This indicates that, under this parameter setting, the likelihood of a certain state occurring within the system is relatively high, but the specific value fluctuates within this interval. Fig. 20(b) points out that when the parameter β_J reaches its critical value of 2, the peak value of the probability density function is exactly 15 000. This means that, at this specific parameter setting, the probability of a certain event occurring within the model reaches a particularly high level. Fig. 20(c) indicates that when the parameter β_J is set to 3, the peak value of the probability density function lies within the broader interval of 15,000 to 20,000. This suggests that, under this parameter setting, the likelihood of a certain state occurring within the system is still relatively high, but more variable or uncertain compared to the situation in the critical case (i.e., when β_J).

As the phase space point (x, v) changes over time, three-dimensional simulations reveal a single prominent peak in the probability distribution $p(x, v, t)$ governed by the Fokker–Planck equation (4.17) with parameters $\kappa = 0.6965$ and $\beta_J = 2$, specifically for a fixed time $t = 0.001$. Here's how the peak's height varies based on different values of $\sqrt{2D}$. For $\sqrt{2D} = 0.3$, the peak reaches a height of 2.5×10^4 in probability density, as exemplified in Fig. 21(a). At the critical threshold where $\sqrt{2D} = 0.6$, the peak's height diminishes to 12 000, as suggested by Fig. 21(b). When $\sqrt{2D}$ increases to 0.9, the peak's height decreases further to 8000 in probability density, as suggested by Fig. 21(c). In summary, it was discovered that as noise intensity increases, the peak of probability density decreases.

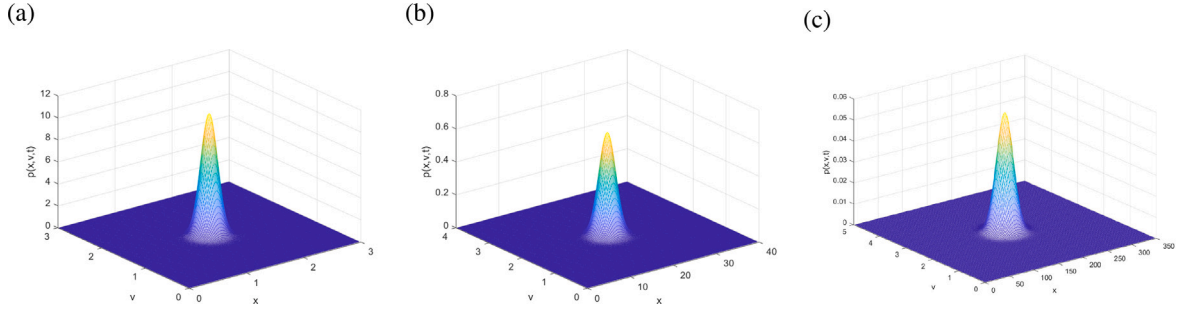


Fig. 19. As the point (x, v) evolves within the phase space, the three-dimensional simulations demonstrate the presence of a solitary peak in the distribution of $p(x, v, t)$ characterized by the Fokker–Planck equation (4.17) with $\beta_J = 1.2$ and $\sqrt{2D} = 0.5$ when considering a constant time value of $t = 10$: (a) When $\kappa = 0.1$, the peak's height of probability density reaches a value of 11; (b) At the critical value of $\kappa = 0.6965$, the peak of probability density experiences a decrease to 0.7; (c) When $\kappa = 2$, the peak's height of probability density further decreases to 0.06.

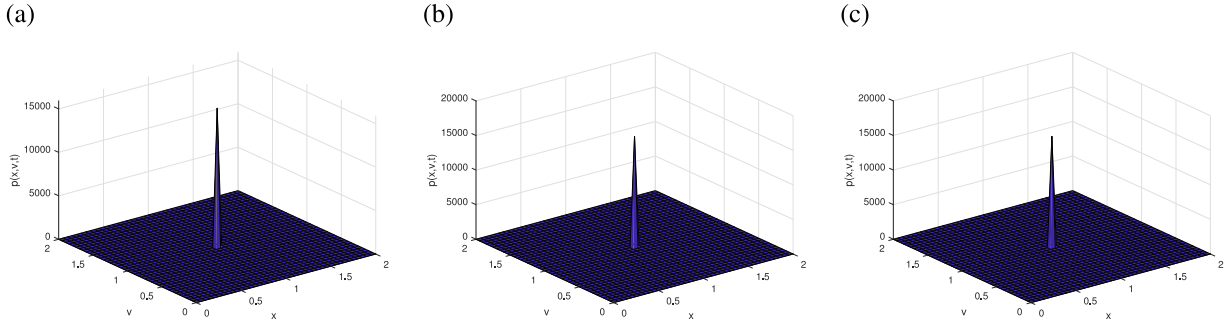


Fig. 20. As the point (x, v) evolves within the phase space, the three-dimensional simulations demonstrate the presence of a solitary peak in the distribution of $p(x, v, t)$ characterized by the Fokker–Planck equation (4.17) with $\kappa = 0.6965$ and $\sqrt{2D} = 0.5$ when considering a constant time value of $t = 0.001$: (a) When $\beta_J = 1$, the peak's height of probability density falls within the range of 10000 to 15000; (b) At the critical value of $\beta_J = 2$, the peak of probability density equals 15000; (c) When $\beta_J = 3$, the peak's height of probability density falls within the range of 15000 to 20000.

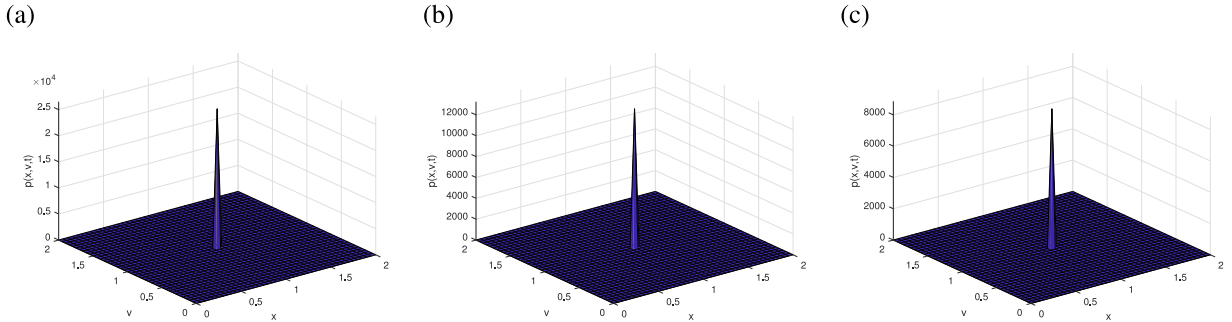


Fig. 21. As the point (x, v) evolves within the phase space, the three-dimensional simulations demonstrate the presence of a solitary peak in the distribution of $p(x, v, t)$ characterized by the Fokker–Planck equation (4.17) with $\kappa = 0.6965$ and $\beta_J = 2$ when considering a constant time value of $t = 0.001$: (a) When $\sqrt{2D} = 0.3$, the peak's height of probability density reaches a value of 2.5×10^4 ; (b) At the critical value of $\sqrt{2D} = 0.6$, the peak of probability density experiences a decrease to 12000; (c) When $\sqrt{2D} = 0.9$, the peak's height of probability density further decreases to 8000.

5. Mean first passage time and noise enhanced stability

To derive the mean first passage time (MFPT) for the resistively shunted superconducting tunnel junction described by the stochastic model (3.11), we consider the high-damping limit ($\beta_J \gg 1$), where the inertia term $\ddot{x}$ becomes negligible. This reduces the system to the overdamped Langevin equation

$$\beta_J \dot{x} = \kappa - \sin x + \sqrt{2D}\eta(t). \quad (5.18)$$

The potential $\Psi(x)$ is defined such that the deterministic force is $-\partial\Psi/\partial x$:

$$\frac{\partial\Psi}{\partial x} = \sin x - \kappa \Rightarrow \Psi(x) = -\cos x - \kappa x + C, \quad (5.19)$$

where C is a constant (set to 0 without loss of generality). Thus, $\Psi(x) = -\cos x - \kappa x$.

The MFPT $\mathbb{E}\tau$ from an initial position x_0 to the barrier at $x_{\max}$ (the location of the nearest local maximum to the right of x_0) is derived from the backward Kolmogorov equation

$$\left[v \frac{\partial}{\partial x} + \left(-\beta_J v - \frac{\partial\Psi}{\partial x} \right) \frac{\partial}{\partial v} + D \frac{\partial^2}{\partial v^2} \right] \mathbb{E}\tau = -1,$$

where the boundary conditions are absorbing at $x_{\max}$ and reflecting at $x \rightarrow -\infty$. The general solution for the MFPT is

$$\text{MFPT}(x_0) = \frac{1}{D} \int_{x_0}^{x_{\max}} \exp\left(-\frac{\Psi(x)}{k_B T}\right) \left[\int_{-\infty}^x \exp\left(\frac{\Psi(x')}{k_B T}\right) dx' \right] dx. \quad (5.20)$$

This equation calculates the average time for a randomly moving particle (e.g., a Brownian particle) to travel from an initial position x_0 to a target position $x_{\max}$ in a potential energy landscape $\Psi(x)$, subject

to thermal fluctuations at temperature T . The inner integral

$$I_{\text{inner}}(x) = \int_{-\infty}^x \exp\left(\frac{\Psi(x')}{k_B T}\right) dx'$$

computes a weighted accumulation from $-\infty$ to x . It represents the effective resistance to reach x from the potential well. The term $\exp(\Psi(x')/k_B T)$ enhances contributions from low-energy regions (deeper in the potential well). The requirement $\Psi(x') \rightarrow \infty$ as $x' \rightarrow -\infty$ ensures convergence (e.g., in a confining potential well). The outer integral

$$I_{\text{outer}} = \int_{x_0}^{x_{\text{max}}} \exp\left(-\frac{\Psi(x)}{k_B T}\right) \cdot I_{\text{inner}}(x) dx$$

weights the inner integral by $\exp\left(-\frac{\Psi(x)}{k_B T}\right)$ (Boltzmann factor). It sums the escape time contributions across the barrier, weighted by thermal statistics. It suppresses high-energy regions (e.g., near barriers) and integrates from x_0 to x_{max} . It is multiplied by $1/D$. Lower diffusivity D leads to longer escape times.

Substituting $\Psi(x) = -\cos x - \kappa x$ and $D = \beta_J k_B T$ into (5.20) yields

$$\begin{aligned} \text{MFPT}(x_0) &= \frac{1}{\beta_J k_B T} \int_{x_0}^{x_{\text{max}}} \exp\left(-\frac{-\cos x - \kappa x}{k_B T}\right) \\ &\times \left[\int_{-\infty}^x \exp\left(\frac{-\cos x' - \kappa x'}{k_B T}\right) dx' \right] dx. \end{aligned} \quad (5.21)$$

Simplifying the exponentials:

$$\begin{aligned} \exp\left(-\frac{-\cos x - \kappa x}{k_B T}\right) &= \exp\left(\frac{\cos x + \kappa x}{k_B T}\right), \quad \exp\left(\frac{-\cos x' - \kappa x'}{k_B T}\right) \\ &= \exp\left(-\frac{\cos x' + \kappa x'}{k_B T}\right), \end{aligned}$$

the expression becomes

$$\begin{aligned} \text{MFPT}(x_0) &= \frac{1}{\beta_J k_B T} \int_{x_0}^{x_{\text{max}}} \exp\left(\frac{\cos x + \kappa x}{k_B T}\right) \\ &\times \left[\int_{-\infty}^x \exp\left(-\frac{\cos x' + \kappa x'}{k_B T}\right) dx' \right] dx. \end{aligned} \quad (5.22)$$

The location of the barrier is given by $x_{\text{max}} = \pi - \arcsin(\kappa)$ for the parameter range $0 < \kappa < 1$. This point represents the first local maximum to the right of the well located at $x_{\text{min}} = \arcsin(\kappa)$. Regarding convergence properties, as $x \rightarrow -\infty$, the function $\Psi(x) = -\cos x - \kappa x$ diverges to infinity (for $\kappa > 0$); this asymptotic behavior is essential as it ensures the convergence of the associated inner integral. Concerning the dynamics, an overdamped approximation is valid under conditions of high damping, specifically when $\beta_J \gg 1$. In contrast, for the underdamped case, no closed-form analytical solution exists, and Kramers' asymptotic formula is typically employed instead.

To explore noise enhanced stability (NES) in the resistively shunted superconducting tunnel junction, we analyze the stochastic model (3.11). NES occurs when an optimal noise intensity D increases the mean escape time from a metastable state, counterintuitively enhancing stability.

For the potential $\Psi(x) = -\cos x - \kappa x$ with a bias current ratio $\kappa < 1$, metastable states arise at specific minima at which the system can become temporarily trapped. These minima occur at positions $x_{\text{min}} = \arcsin(\kappa) + 2\pi n$, where n is an integer indexing the periodic wells of the potential. At these points, the force $-\partial\Psi/\partial x = \kappa - \sin x$ vanishes, confirming extrema, and the second derivative $\partial^2\Psi/\partial x^2 = \cos x$ is positive since $\cos(\arcsin(\kappa)) = \sqrt{1-\kappa^2} > 0$ for $\kappa < 1$. Thus, each x_{min} corresponds to a local energy minimum. Due to the periodicity of the cosine term and the linear tilt $-\kappa x$, these metastable states repeat every 2π along x , forming a series of wells where the system may reside before escaping via thermal or quantum fluctuations. The escape from a metastable minimum requires surmounting a potential barrier, with the nearest peak located at $x_{\text{max}} = \pi - \arcsin(\kappa)$. Here, the force $\kappa - \sin x$ also vanishes, but the second derivative $\cos(\pi - \arcsin(\kappa)) = -\sqrt{1-\kappa^2} < 0$

confirms a local maximum. The height of the escape barrier $\Delta\Psi$ is the potential difference $\Psi(x_{\text{max}}) - \Psi(x_{\text{min}})$, which governs transition rates via Arrhenius-like behavior. As $\kappa \rightarrow 1^-$, both x_{min} and x_{max} approach $\pi/2$, causing the barrier to vanish and the metastability to break down. For $\kappa > 1$, the tilt dominates, and no local minima exist, highlighting the critical role of $\kappa < 1$ in defining metastable regimes.

The equations of motion in first-order form, given by (4.15), are numerically integrated using the Euler–Maruyama method. This approach discretizes the dynamics into time steps of size Δt , where the position x and velocity v evolve as follows:

$$\begin{aligned} x_{n+1} &= x_n + v_n \Delta t, \\ v_{n+1} &= v_n + (\kappa - \beta_J v_n - \sin x_n) \Delta t + \sqrt{2D\Delta t} \xi_n. \end{aligned}$$

The position update $x_{n+1} = x_n + v_n \Delta t$ is deterministic, advancing the state using the current velocity v_n . In contrast, the velocity update $v_{n+1} = v_n + (\kappa - \beta_J v_n - \sin x_n) \Delta t + \sqrt{2D\Delta t} \xi_n$ incorporates both deterministic forces and stochastic noise. Here, κ represents a constant drive, $-\beta_J v_n$ denotes linear damping, and $-\sin x_n$ arises from a periodic potential, while the term $\sqrt{2D\Delta t} \xi_n$ adds Gaussian noise scaled by the diffusion coefficient D , with $\xi_n \sim \mathcal{N}(0, 1)$ being independent standard normal random variables at each step. This method efficiently captures the combined deterministic and stochastic dynamics of the system.

The MFPT behavior in Fig. 22(a) reveals distinct noise-dependent regimes. At very low noise intensities, the MFPT approaches T_{max} due to rare escape events from the metastable state. For $D > 0.3$, the MFPT decreases exponentially, aligning with the expected Arrhenius law dependence where higher noise facilitates escape. Notably, a subtle non-monotonic peak in MFPT emerges within the intermediate noise range of $D \in [0.15, 0.25]$, indicative of a possible weak NES effect. This NES phenomenon manifests as a stabilization of the metastable state by intermediate noise levels, counterintuitively increasing the escape time compared to slightly lower or higher noise. This stabilization arises from resonant interactions between the noise and the intrinsic oscillations within the potential well. The NES trend aligns with classical metastable systems investigated by Spagnolo, Fiasconaro and Dubkov [68].

To enhance the observability and strength of the NES peak in Fig. 22(b), we increase the damping coefficient to $\beta_J = 0.5$ to better reveal the resonant interaction; reduce the potential bias to $\kappa = 0.2$ to decrease the inherent escape tendency; extend the maximum simulation time to $T_{\text{max}} = 5000$ to adequately capture long escape events at low noise; increase the number of realizations to $N_{\text{realizations}} = 500$ to improve statistical reliability of the MFPT estimates, particularly in the peak region; and focus the noise intensity range more densely on $D \in [0.05, 0.3]$ to precisely resolve the NES peak characteristics. The system (4.15) exhibits a metastable state at $x_{\text{min}} = \arcsin(\kappa) \approx 0.2014$, separated from an unbound state by an escape barrier at $x_{\text{max}} = \pi - \arcsin(\kappa) \approx 2.9402$. Simulation results for the MFPT reveal distinct dynamical regimes based on noise intensity D . At low noise ($D < 0.15$), escape events are exceedingly rare within the maximum simulation time $T_{\text{max}} = 5000$. Consequently, most trajectories do not escape, resulting in MFPT values clustered near 5000. For moderate noise ($0.15 \leq D \leq 0.25$), the MFPT decreases monotonically as D increases, indicating that higher thermal energy facilitates more frequent escapes over the barrier. At high noise ($D > 0.25$), escapes occur rapidly, driving the MFPT toward small values. This behavior is reflected in the plot of MFPT versus D , which shows a consistent monotonic decrease in escape time with increasing noise intensity, highlighting the inverse relationship between thermal fluctuations and the stability of the metastable state.

To visualize the dynamics of the physics model (4.15), we simulate a single trajectory driven by noise in (x, v) space, as shown in Fig. 23(a). The simulation begins at the stable fixed point $x = \arcsin(\kappa) \approx 0.5236$, with zero initial velocity. Termination occurs if the path escapes by reaching $x \geq \pi - \arcsin(\kappa) \approx 2.618$; this escape boundary is marked by a dashed line. Otherwise, the simulation halts at the maximum time

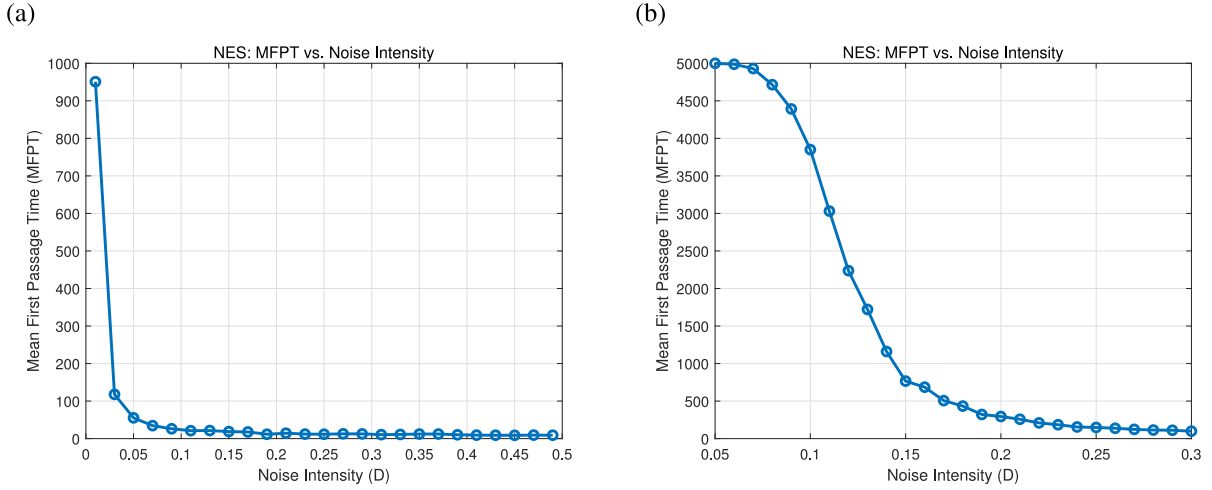


Fig. 22. It simulates the noise enhanced stability effect in the Josephson junction system (4.15) with the following parameter settings: (a) damping coefficient $\beta_J = 0.1$, bias current ratio $\kappa = 0.5$, noise intensity D ranging from 0.01 to 0.5, time step $dt = 0.01$, maximum simulation time $T_{\max} = 1000$, and $N_{\text{realizations}} = 100$; (b) damping coefficient $\beta_J = 0.5$, bias current ratio $\kappa = 0.2$, noise intensity D ranging from 0.05 to 0.3, time step $dt = 0.01$, maximum simulation time $T_{\max} = 5000$, and $N_{\text{realizations}} = 500$.

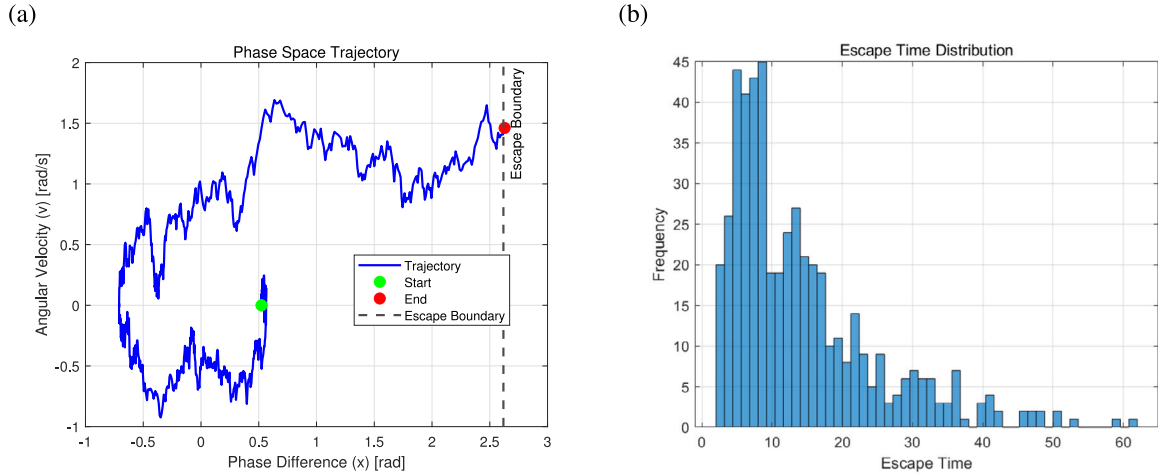


Fig. 23. (a) We simulate a single trajectory in (x, v) space and visualizes its dynamics. The simulation includes noise diffusion and terminates if the phase difference exceeds an escape boundary. Parameters: damping coefficient $\beta_J = 0.1$, bias current ratio $\kappa = 0.5$, noise intensity $D = 0.2$, time step $dt = 0.01$, and maximum simulation time $T_{\max} = 500$; (b) It simulates the escape time of a particle from a potential well using Langevin dynamics in system (4.15). The parameters are set to $\beta_J = 0.1$, $\kappa = 0.5$ and $D = 0.2$, with a time step $dt = 0.01$ and a maximum simulation time $T_{\max} = 1000$. The simulation runs for $N_{\text{realizations}} = 500$ independent realizations. The particle starts at the stable fixed point $x = \arcsin(\kappa)$ and escapes when it reaches $x_{\max} = \pi - \arcsin(\kappa)$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$T_{\max} = 500$ s. The trajectory path (with a green marker for the start and red for the end) illustrates how stochastic noise drives the system out of its initial metastable state. Escape, which signifies a transition past 150° , models the behavior observed in Josephson junction system (4.15), demonstrating the stochastic dynamics of transitions between metastable states.

The histogram in Fig. 23(b), displaying the distribution of escape times across 50 bins, exhibits the expected asymmetric, exponential-like shape characteristic of an escape process driven by thermal activation over a potential barrier. Given the simulation parameters, nearly all realizations ($\approx 100\%$) escape within the maximum time $T_{\max}$, and non-escape events (where $t \geq T_{\max}$ without escaping) are filtered out; consequently, the histogram shows only successful escape times. The average escape time in this distribution depends critically on the barrier height and the noise intensity D . Computationally, the escape dynamics are simulated using a time step $dt = 0.01$, chosen for stability. The potential well boundaries $x_{\min}$ and $x_{\max}$, which define the escape condition ($x \geq x_{\max}$), are calculated based on the parameter κ . At each step, thermal noise with variance scaled by $\sqrt{2Ddt}$ is applied.

The velocity update combines damping ($-\beta_J v$), the force term derived from the potential ($\kappa - \sin(x)$), and the scaled noise term, while the position is updated via Euler integration ($x = x + vdt$). The simulation loop terminates early upon escape, recording the escape time.

Through optimized simulations employing vectorization (parallel computation of all realizations per (β_J, D) pair), early termination upon escape, and periodic boundary handling via modulo 2π to confine position x , key observations emerge on how damping and noise intensity influence the MFPT for a particle escaping from $x_{\min} = \arcsin(\kappa)$ to $x_{\max} = \pi - \arcsin(\kappa)$ as shown in Fig. 24(a). For low damping ($\beta_J = 0.05$), the underdamped system exhibits resonant behavior, resulting in a non-monotonic MFPT hump at moderate noise intensities (D). Here, noise synchronizes with the particle's natural oscillations, temporarily trapping it in the potential well and increasing escape times. In contrast, at higher damping ($\beta_J = 0.1$ or 0.2), the NES effect diminishes as damping increases, flattening the MFPT curves and yielding a monotonic decrease in escape times with D ; strong damping suppresses oscillations, reducing the resonance underpinning NES. MFPT behavior is further modulated by D : at small D , escape is

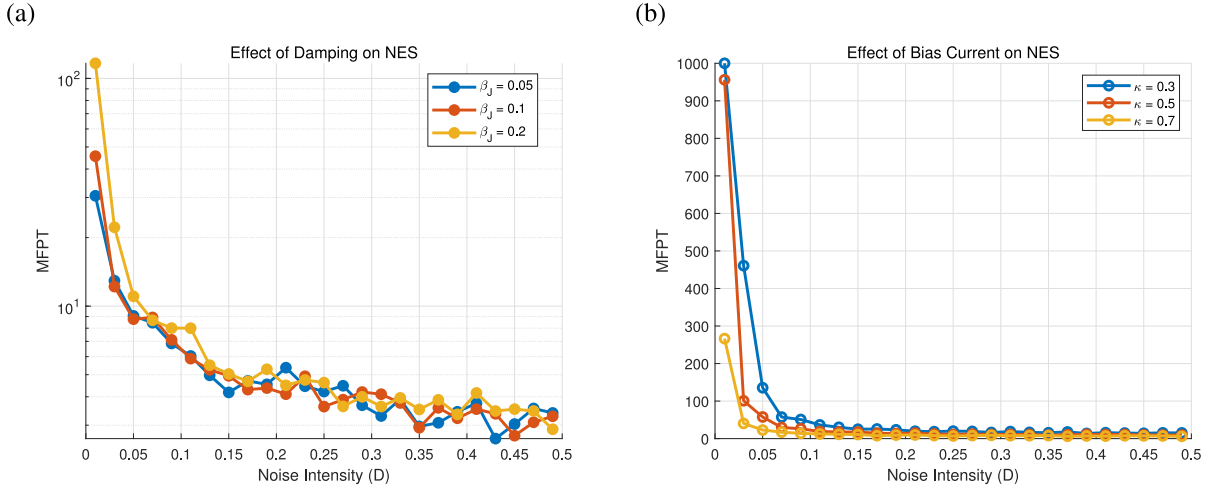


Fig. 24. (a) This simulation models the escape process of a particle in a tilted periodic potential from system (4.15), calculating the MFPT for different damping coefficients (β_J) and noise intensities (D). The particle starts at the potential minimum ($x_{\min} = \arcsin(\kappa)$) and escapes when it crosses the barrier at $x_{\max} = \pi - x_{\min}$. Numerical integration of the stochastic differential equations uses the Euler-Maruyama method. Parameter settings: damping coefficient $\beta_J = [0.05, 0.1, 0.2]$, bias current ratio $\kappa = 0.5$, noise intensity $D \in [0.01, 0.5]$, time step $dt = 0.01$, maximum simulation time $T_{\max} = 1000$, and $N_{\text{realizations}} = 100$; (b) It simulates the MFPT for a particle escaping from a potential well under the influence of noise and a bias current. Parameter settings: damping coefficient $\beta_J = 0.1$, bias current ratio $\kappa = [0.3, 0.5, 0.7]$, noise intensity D varying from 0.01 to 0.5, time step $dt = 0.01$, maximum simulation time $T_{\max} = 1000$, and number of realizations $N_{\text{realizations}} = 100$.

rare (high MFPT) due to inadequate energy to overcome the barrier, while at large D , escape is rapid (low MFPT) as noise dominates and readily propels the particle over the barrier. The NES effect peaks at moderate D , particularly under weak damping, where noise and system dynamics synergistically delay escape. This non-monotonicity arises from the interplay of noise (injecting energy), damping (dissipating energy and quenching oscillations), and the potential tilt (biasing motion toward $x_{\max}$), aligning with extensions of Kramers' escape theory to underdamped regimes across the tested bias current $\kappa = 0.5$ and noise intensities ($D = 0.01$ to 0.5). Augello, Valenti, Pankratov, et al. [69] observed a non-monotonic behavior of the lifetime as a function of the current driving signal frequency and noise correlation time. Moreover, they found two maxima in the mean escape time's non-monotonic dependence on correlated noise intensity. Quantum analogs in [70] showed similar non-monotonic MFPT vs. D behavior, but with tunneling corrections.

The key aspects of the system (4.15), illustrated in Fig. 24(b), include its dynamics, escape conditions, and parameters. The particle's motion starting from a local minimum is governed by Langevin equations that incorporate damping and a sinusoidal potential with a bias current κ . The plot depicts the MFPT versus noise intensity (D) for varying bias currents (κ). Escape time from the metastable state depends critically on both the energy barrier height and D . A lower κ value increases the energy barrier, generally resulting in longer escape times as more energy is needed to overcome it. Conversely, a higher diffusion coefficient D (greater noise intensity) typically shortens escape times by supplying more frequent or energetic kicks to propel the particle over the barrier. However, the phenomenon of NES reveals a counterintuitive effect: at intermediate D levels, the escape time can peak, becoming longer than at either very low or very high noise intensities. This peak arises from resonant trapping, where the noise frequency constructively interferes with the particle's oscillatory motion within the well, effectively confining it for extended periods. Crucially, this resonant trapping effect is significantly more pronounced and observable when the energy barrier is higher, corresponding to lower values of κ .

We compute the potential $\Psi(x) = -\kappa x - \cos(x)$ with a tilt parameter $\kappa = 0.5$, and plot it versus $x \in [-\pi, 3\pi]$ (using $x = \text{linspace}(-\pi, 3\pi, 1000)$ to generate 1000 points) in Fig. 25(a). The blue curve represents $\Psi(x)$. This potential, exhibiting an overall decrease due to the linear term $-\kappa x$, creates a washboard shape. A metastable state (a local minimum

where particles can be trapped) occurs at $x_{\min} = \arcsin(\kappa) = \pi/6 \approx 0.5236$, while the escape barrier (a local maximum representing the energy barrier for escape) is at $x_{\max} = \pi - \arcsin(\kappa) = 5\pi/6 \approx 2.618$. For visibility, a red dashed vertical line marks $x_{\min}$ and a green dashed vertical line marks $x_{\max}$, extending from $\Psi = -4$ to $\Psi = 0$. The actual potential values are $\Psi(x_{\min}) \approx -1.1278$ and $\Psi(x_{\max}) \approx -0.4430$. Although additional critical points exist at $x = \pi/6 + 2\pi$ (minimum) and $x = 5\pi/6 + 2\pi$ (maximum) within the interval $[\pi, 3\pi]$, only the primary pair ($x_{\min}$ and $x_{\max}$) is explicitly marked in the figure.

The simulation begins by establishing key physical parameters, including the damping coefficient β_J , the driving force amplitude κ , and the noise intensity D . The time step is set to dt . The total duration is $T_{\max}$. Initial conditions place the system (4.15) at $x = \arcsin(\kappa)$ with zero initial velocity ($v = 0$), while the escape barrier is defined at $x = \pi - \arcsin(\kappa)$. During each iteration, the current phase difference x and time t are recorded. The velocity is updated via Euler-Maruyama integration, incorporating the deterministic force and stochastic noise. Subsequently, the position x is advanced using the new velocity. The loop terminates if x surpasses the escape barrier or if t reaches $T_{\max}$. Finally, the trajectory is plotted with the escape barrier highlighted as a dashed red line for visual reference, as shown in Fig. 25(b).

The plot of MFPT versus noise intensity (D) in Fig. 26(a) exhibits a distinct peak at $D \approx 0.12$, demonstrating the NES phenomenon where moderate noise paradoxically increases the escape time, thereby stabilizing the system (4.15). This peak arises from the resonant coupling of noise with the system dynamics, effectively trapping the particle longer within the potential well. At low D values, noise provides insufficient energy to overcome the barrier, while high D dominates the dynamics and reduces escape time. Blue circles represent the simulated MFPT for each D , calculated from stochastic trajectories integrated using the Euler-Maruyama method. The escape threshold was defined from $x_{\min} = \arcsin(\kappa)$ to $x_{\max} = \pi - x_{\min}$. Realizations not escaping within the maximum simulation time ($T_{\max}$) contributed $T_{\max}$ to the MFPT, potentially underestimating the true MFPT at low D . Overall, the simulation successfully captures NES, highlighting noise-induced stabilization and revealing that optimal stability for this metastable system occurs near $D \approx 0.12$ (with slight variations expected due to stochasticity). Gordeeva, Pankratov and Spagnolo [71] presented the analysis of the mean switching time and its standard deviation in short overdamped Josephson junctions, driven by a direct current and a periodic signal, and found that fluctuations may both decrease and increase the switching time.

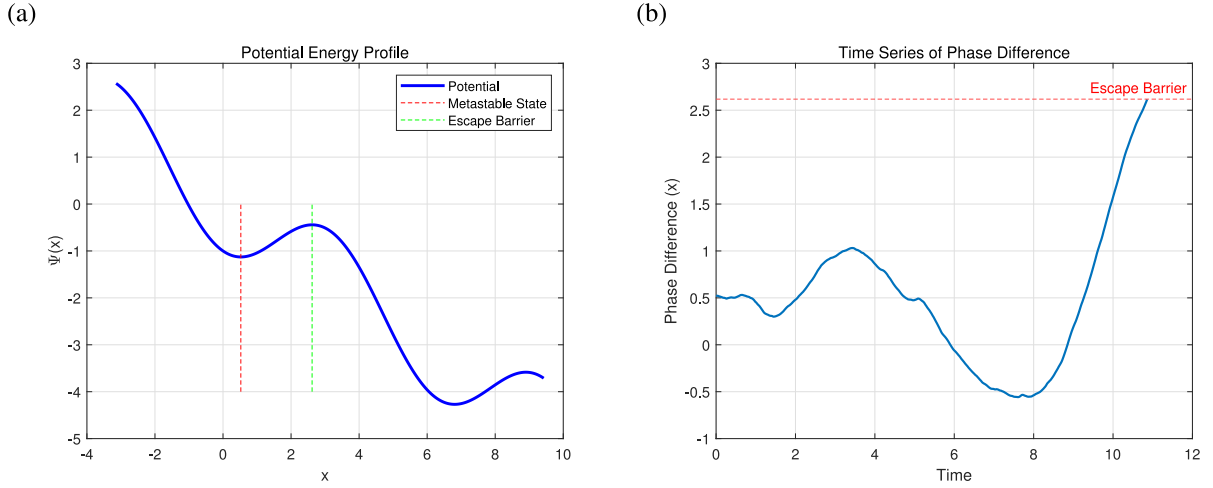


Fig. 25. (a) It plots the potential energy function $\Psi(x) = -\kappa x - \cos(x)$ for $\kappa = 0.5$ over the interval $x \in [-\pi, 3\pi]$; (b) It captures the stochastic dynamics of the system (4.15), including the escape condition occurring when the phase difference surpasses the critical barrier. Parameters are set as follows: damping coefficient $\beta_J = 0.1$, bias current ratio $\kappa = 0.5$, noise intensity $D = 0.2$, time step $dt = 0.01$, and maximum simulation time $T_{\max} = 100$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

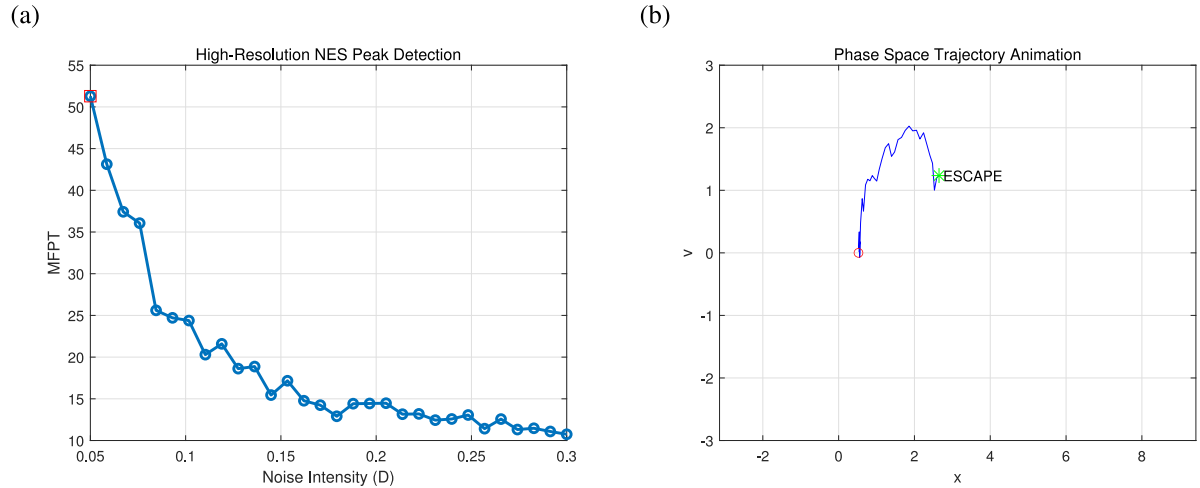


Fig. 26. (a) It simulates a bistable system to investigate NES by computing the MFPT for a particle escaping from a potential well under varying noise intensities D . The system dynamics follow a damped Eq. (4.15) with stochastic noise. Key parameters are the damping coefficient $\beta_J = 0.1$, driving force $\kappa = 0.5$, and D values ranging from 0.05 to 0.3 (30 points). The simulation runs with 200 realizations per D value, using a time step $dt = 0.005$ and a maximum time $T_{\max} = 2000$; (b) The solution involves simulating the particle's motion under the given stochastic differential Eq. (4.15) and animating its path in phase space until it escapes the metastable state. Parameter settings: damping coefficient $\beta_J = 0.1$, bias current ratio $\kappa = 0.5$, noise intensity $D = 0.2$, time step $dt = 0.05$, and maximum simulation time $T_{\max} = 100$.

To animate the trajectory of Eq. (4.15) in phase space, the physical parameters (β_J , κ , D) and simulation settings (dt , $T_{\max}$) are first defined. Initial conditions are set starting from a metastable state located at $x_{\min} = \arcsin(\kappa)$. The particle's trajectory is then precomputed using the Euler-Maruyama method for stochastic integration, where noise is incorporated by adding a random normal variable scaled by $\sqrt{2Ddt}$. The simulation loop terminates either if the particle escapes the metastable well, defined as reaching $x \geq \pi - \arcsin(\kappa)$, or if the simulation time exceeds $T_{\max}$. Within the resulting figure, the metastable state is marked with a red dot. If an escape occurs, a green star and the text "ESCAPE" are displayed at the exit point. The animation runs at approximately 20 frames per second (corresponding to a 50 ms interval between frames), producing a visualization of the particle's stochastic path in phase space and highlighting its escape from the metastable state when it happens, as in Fig. 26(b).

Fig. 27 focuses on the low-noise regime (D ranging from 0.01 to 0.1 in steps of 0.005) to capture long escape times for accurate low-noise MFPT measurements. Five initial positions span the metastable well from $x_{\min}$ to $x_{\max} - 0.05$. The central semilog plot shows MFPT versus

D with exponential fits for all positions, revealing exponential scaling ($\text{MFPT} \propto \exp(\Delta\Psi/D)$). The slopes steepen for positions farther from the barrier. Fits are excellent except near saturation ($\text{MFPT} \approx T_{\max}$). Physical insights emerge from three insets: a tilted washboard potential (top-left) marking positions; exponential fits (bottom-right) extracting effective activation energies that decrease as the initial position moves toward the barrier; and $\Delta\Psi$ versus x_0 (top-right) showing barrier reduction from ~ 1.25 near $x_{\min}$ to ~ 0.5 at $x_{\max} - 0.05$. This 50%–60% reduction in $\Delta\Psi$ quantitatively explains orders-of-magnitude decreases in MFPT and confirms theoretical limits: the maximum $\Delta\Psi$ approaches the calculated barrier height $\Delta\Psi_{\text{theoretical}}^{\text{theoretical}} \approx 1.25$ for $\kappa = 0.5$. Crucially, positions near the well bottom $x_{\min}$ exhibit the strongest noise dependence, while barrier-proximal starts enable operation at lower noise intensities. Collectively, this multi-panel visualization quantifies how initial position affects noise-activated escape in the low-noise regime, connecting the potential landscape, MFPT behavior, and activation energy physics. The quantitative outputs offer concrete design guidelines for Josephson junction applications.

Fig. 28 presents simulated MFPT ratios relative to the baseline position, investigating noise-activated escape dynamics in system (3.11).

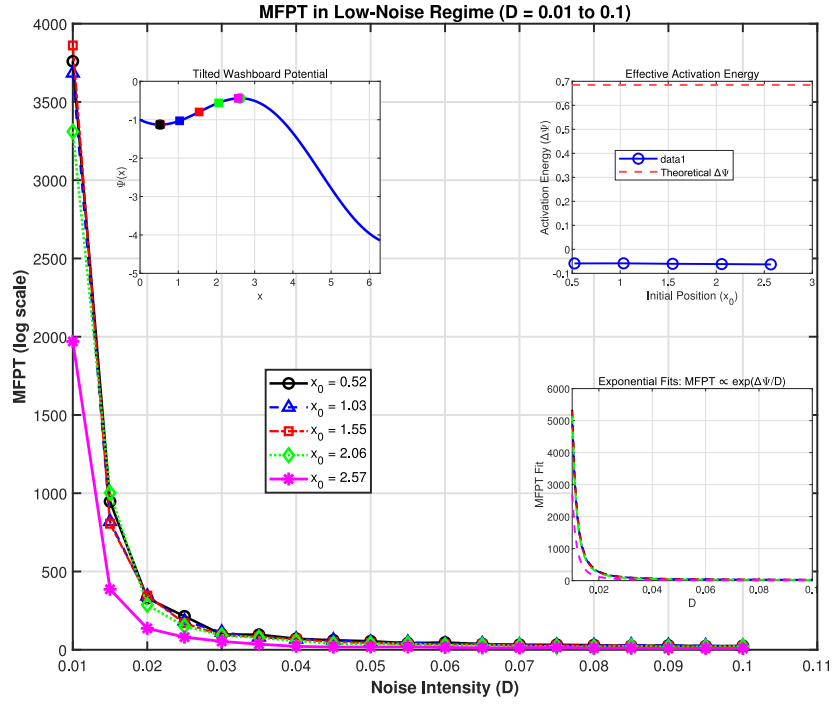


Fig. 27. Simulation parameters for analyzing the MFPT behavior of system (3.11) in the low-noise regime across different initial positions are set as follows: damping coefficient $\beta_f = 0.1$, bias current ratio $\kappa = 0.5$, noise intensity $D \in [0.01, 0.1]$, time step $dt = 0.01$, and maximum simulation time $T_{\max} = 5000$ (extended to capture rare escape events). The number of trajectories is $N_{\text{realizations}} = 100$. Metastable states are defined by the potential minimum at $x_{\min} = \arcsin(\kappa)$ and escape barrier at $x_{\max} = \pi - x_{\min}$. Five initial positions spanning the well from $x_{\min}$ to $x_{\max} - 0.05$ are used.

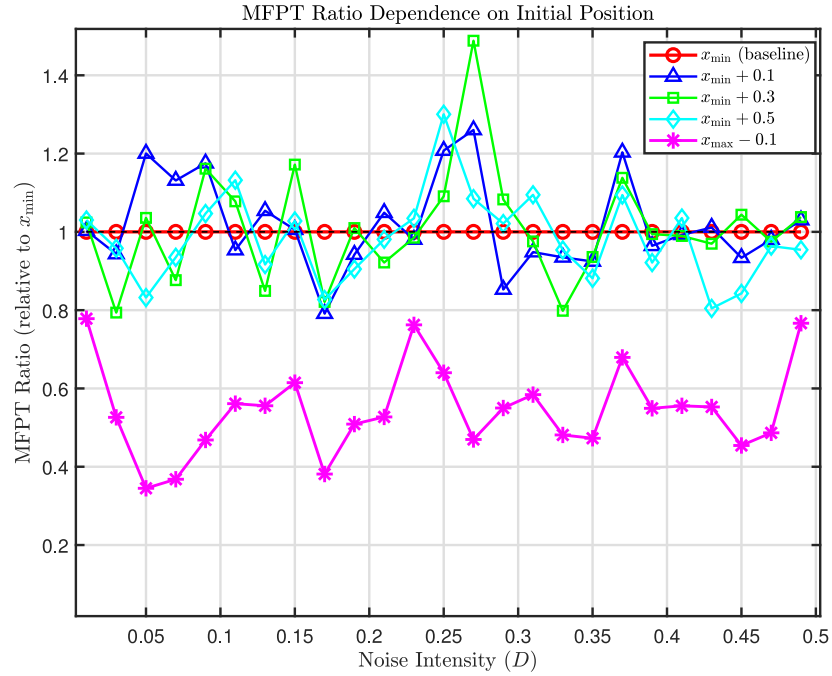


Fig. 28. The MFPT ratios are calculated relative to the baseline (initial position at the potential minimum). Parameters: damping coefficient $\beta_f = 0.1$, bias current ratio $\kappa = 0.5$, noise intensity range $D \in [0.01, 0.5]$, time step $dt = 0.01$, max simulation time $T_{\max} = 1000$, and number of trajectories $N_{\text{realizations}} = 100$. Metastable states are defined by the potential minimum $x_{\min} = \arcsin(\kappa)$ and escape barrier $x_{\max} = \pi - x_{\min}$. Initial positions tested: baseline position (potential minimum), $x_{\min}$; slightly right of minimum, $x_{\min} + 0.1$; mid-well position, $x_{\min} + 0.3$; higher in well, $x_{\min} + 0.5$; and near the escape barrier, $x_{\max} - 0.1$.

Visual enhancements include position-specific markers. The ratio of MFPT to the baseline value at $x_{\min}$ quantifies escape efficiency: unity indicates equivalent escape time, values below 1 show acceleration (e.g., ratios of 0.05–0.5 for near-barrier starts at $x_{\max} - 0.1$), while values above 1 denote slowdown. Dramatic MFPT reduction occurs

for barrier-proximal starts at low noise ($D < 0.1$), yielding faster escape than baseline. Near-barrier starts $x_{\max} - 0.1$ reduce MFPT by up to 70%–90% at low noise. Mid-well positions ($x_{\min} + 0.3$ and $x_{\min} + 0.5$) exhibit moderate reduction (ratios 0.5–0.9), diminishing with increasing noise. Position sensitivity dominates below $D < 0.3$

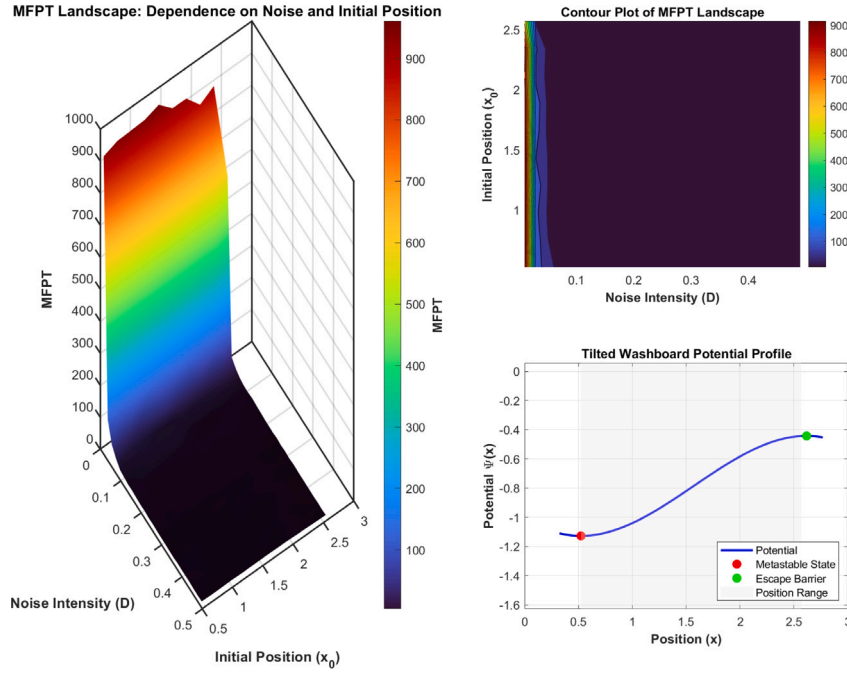


Fig. 29. The 3D surface plot depicts the MFPT as a function of noise intensity D and initial position x_0 . Parameter settings include: damping coefficient $\beta_J = 0.1$, bias current ratio $\kappa = 0.5$, noise intensity range $D \in [0.01, 0.5]$, time step $dt = 0.01$, maximum simulation time $T_{\max} = 1000$, and $N_{\text{realizations}} = 100$ trajectories. The potential minimum is defined by $x_{\min} = \arcsin(\kappa)$, the escape barrier by $x_{\max} = \pi - x_{\min}$, and simulations used 10 initial positions spanning $x_{\min}$ to $x_{\max} - 0.05$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

but vanishes at higher noise where ratios converge to 1 as thermal energy homogenizes dynamics. At $D > 0.3$, position has minimal effect ($<10\%$ reduction). Position sensitivity decreases with increasing noise. Critically, barrier proximity reduces the noise energy required for escape, enabling faster switching in applications like Josephson junction devices. Maximum reduction occurs at lowest noise ($D \approx 0.01$), which is most beneficial for low-noise applications (cryogenic systems). The ratio visualizations demonstrate how strategic initial positioning significantly optimizes metastable escape in noise-constrained systems, providing direct performance metrics particularly in low-noise operating regimes.

The comprehensive visualization in Fig. 29 employs a multi-panel layout to explore metastable escape dynamics, anchored by an interactive 3D surface plot mapping MFPT against noise intensity D and initial position x_0 . Utilizing 10 initial positions for surface smoothness and the Turbo colormap for perceptual clarity, the plot features directional lighting, ‘shading interp’ for color interpolation, with annotations highlighting critical regions. Complementary panels include a 2D contour plot revealing MFPT isoclines and a tilted washboard potential profile contextualizing the energy landscape. Interactive elements enable rotation, zoom, and a custom data cursor displaying D , x_0 , MFPT, and barrier height values. The visualization uncovers key physical insights: a global MFPT minimum (fast escape) at high noise intensities with near-barrier starts (top-right), a global maximum (slow escape) at low noise with well-bottom starts (bottom-left), and a transition region where noise and position compete. Contour lines show tradeoffs between parameters, revealing how similar escape times emerge from different $D-x_0$ combinations. Color coding intuitively signals dynamics: blue/purple for rapid escape, yellow/red for prolonged metastability, and green for intermediate durations respectively. By integrating 3D perspectives, quantitative contours, and physical context, this framework elucidates the nuanced interplay between noise-driven activation and initial conditions in barrier-crossing processes.

6. Summary and outlook

In the present work, the equation governing the total junction current was analyzed in accordance with Kirchhoff’s current law. By

converting it to a dimensionless form, the impact of noise was introduced using the fluctuation–dissipation theorem, ultimately leading to the development of a resistively shunted superconducting tunnel junction system perturbed by thermal noise. By setting parameters κ and β_J to zero, the characteristics of the resistively shunted superconducting tunnel junction system were examined, revealing its Hamiltonian nature. The energy surface and phase portrait were simulated to gain further insights. In the presence of non-zero parameters κ and β_J , the system exhibited clockwise hysteresis in its typical I-V characteristic curve. For different parameter values, it was found that a bifurcation phenomenon affecting the planar limit cycle was possible. Furthermore, numerical simulations were conducted to visualize the stochastic dynamics of the system, particularly its response to thermal fluctuations. The Fokker–Planck equation corresponding to the resistively shunted superconducting tunnel junction system under thermal fluctuations was derived. Based on this equation, simulations were conducted to assess the probability density, mean first passage time and noise enhanced stability under specific parameter configurations.

There remains a sizable number of open problems in this direction. Among the significant challenges posed by the experimental observations of charge noise generated by a mesoscopic conductor with a Josephson junction serving as an on-chip detector, we note that the rate at which it switches out of its zero-voltage state is tied to the thermal escape in the presence of non-Gaussian Lévy noises [72,73] that exhibit jumps or discontinuities in its trajectory. Lévy noise, characterized by heavy-tailed distributions, enables rare but large-amplitude jumps, fundamentally altering escape dynamics from metastable states. Unlike Gaussian noise, which primarily induces small perturbations, Lévy noise can trigger abrupt phase slips even at low noise intensities, significantly impacting switching statistics. In the context of Josephson devices, Lévy noise can potentially play a role in describing the dynamics of quantum systems subject to fluctuations or perturbations. Lévy-like noise could arise in Josephson junction from non-thermal sources (e.g., cosmic rays, quasi-particle poisoning) or engineered environments (e.g., superconducting circuits coupled to Lévy baths). The Refs. [50,74–76] may indeed be useful to explore further the potential

impacts of Lévy noise on Josephson devices and other related systems. Dubkov and Spagnolo [77] demonstrated that Lévy-driven systems exhibit anomalous switching rates due to scale-free jump statistics. Dubkov, La Cognata and Spagnolo [78] analyzed the implications of Lévy noise, which formalizes first-passage problems for non-Gaussian processes. For the specific case of Lévy flights with index $\alpha = 1$ and a cubic metastable potential, Dubkov, Guarcello and Spagnolo [79] obtained a closed expression for the mean residence time in quadrature; their analytical results reveal an enhancement of the mean residence time in the metastable state due to the influence of Lévy noise.

It would be extremely interesting to try to identify an effective non-local Fokker–Planck equation, which facilitates the derivation of an explicit expression for the escape rate. Furthermore, the non-zero third moment of current noise may lead to the formation of rate asymmetry. Exploring tunnel junctions exhibiting Lévy noise characteristics is an interesting question in its own right. Identifying these behaviors from the perspective of dynamical systems, as presented herein, and more systematically examining the analysis of experimental data and the optimization of detection circuits would constitute a particularly relevant task for future work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Numerical algorithms source code associated with this article can be found, in the online version, at GitHub.

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